Digital Image Processing

Class Information

Course: Digital Image Processing | NCSH301 | 3-1-0

Class Timing:

Tuesday: 4:00 PM - 4:50 PM

Wednesday : 4:00 PM - 4:50 PM

Thursday: 4:00 PM - 4:50 PM

Friday: 4:00 PM - 4:50 PM (T)

Class Venue: NLHC G-13

Google Classroom link:

https://meet.google.com/rgv-qbei-xhg

Lecture Plan

Lecture Plan

Sl No.	Topic	No. of Hours
1	**Introduction:** Digital Image; Basic Phases; History; Human Visual System; Image Sampling; Quantization; Digital Image Representation; Image Formats; Relationship between Pixels; Pixel Adjacency; Path, Pixels Connectivity; Connected Set; Boundary; Hole; Distance Measurement; Mathematical Operators used in Image Processing.	7
2	**Image Enhancement-I:** Contrast Enhancement; Histogram Processing; Point Processing; Spatial Domain Filtering; Edge Sharpening.	6
3	**Image Enhancement-II:** The Fourier Series; The Fourier Transform; Frequency Domain Filtering: Smoothing Filters and Sharpening Filters; Applications of Correlation; FFT.	3
	Quiz-I (*Syllabus: Sl No. 1-3*)	
4	**Image Restoration:** Noise Models; Noise Smoothing; Linear and Non-linear Filtering; Sharpening; Image Restoration; Motion Blur Removal; Geometric Corrections.	4
	Mid Semester Examination (*Syllabus: Sl No. 1-4*)	

Lecture Plan

Sl No.	Topic	No. of Hours
5	**Image Morphology:** Fundamental Operations; Morphological Algorithms; Applications; Mathematical Examples.	3
6	**Image Segmentation:** Pixel-based Approach; Multilevel Thresholding; Adaptive Thresholding; Optimal Thresholding; Region-based Approach; Point and Line Detection; First and Second Order Edge Operators; Canny Edge Detector; Hough Transform; Edge Linking.	7
7	**Image Compression:** Error Criterion; Lossless Compression: Run-length Coding; Shannon-Fano Coding; Huffman Coding; Arithmetic Coding; Lossy Compression: Block Truncation Compression; Vector Quantization Compression; JPEG Standard.	6
	Quiz-I (*Syllabus: Sl No. 5-7*)	
8	**Image Representation and Description:** Freeman Chain Coding; Binary Tree and Quad Tree Coding; Polygonal Approximation; Boundary Segments; Boundary Extraction; Medial Axis Generation & Thinning; Boundary Descriptors; Regional Descriptors; Topological Descriptors; Relational Descriptors.	3
9	**Multiresolution Analysis and Wavelet:** Pyramidal Coding; Subband Coding; Application of Wavelets.	3
	End Semester Examination (*Syllabus: Sl No. 1-9*)	

Examination Schedule

Quiz – I (Weight : 10%)	Mid Sem (Weight : 30%)	Quiz – II (Weight : 10%)	End Sem (Weight : 50%)
11 Sep, 2026, 4:00 PM – 4:50 PM	As per the academic schedule	13 Nov, 2026, 4:00 PM – 4:50 PM	As per the academic schedule

Text and Reference Books

Text and Reference Books

1. Digital Image Processing, R. C. Gonzalez and R. E. Woods, Pearson Education.
2. Digital Image Processing and Analysis, B. Chanda and D. Dutta Mazumdar, PHI.
3. Fundamentals of Digital Image Processing, A. K. Jain, Pearson India Education.
4. Digital Image Processing, W. K. Pratt, Wiley-Interscience.

Objective of the Course

Objective of the Course

- Provide comprehensive coverage of theory and computation related to image format, image formation, pixel geometry, image processing, and image analysis.

- To provide exposure to morphology, segmentation, compression, and representayion-description techniques applied in this area.

Fundamentals of Digital Image Processing

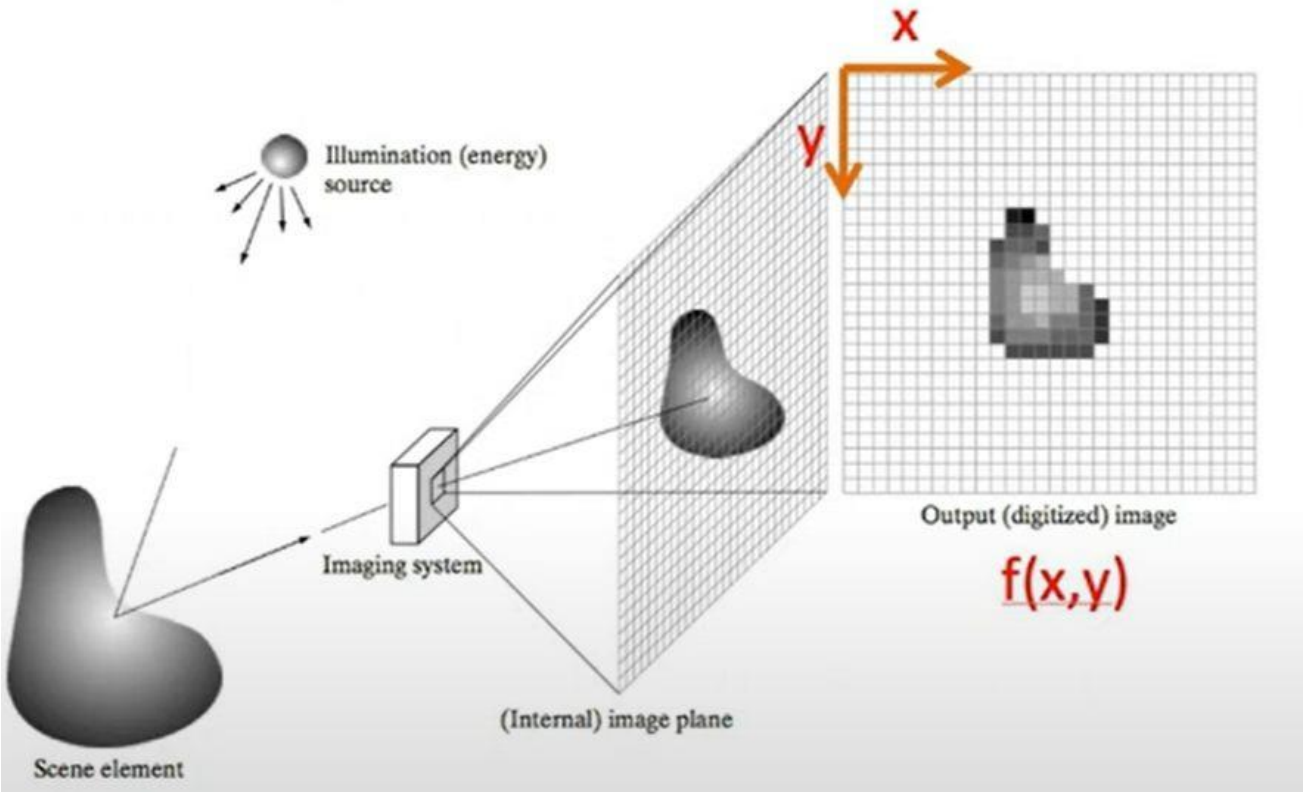
Content derived from multiple sources

Contents

- ✓ What is a Digital Image?
- ✓ Brief History of Digital Image Processing
- ✓ Basic Phases of Digital Image Processing
- ✓ The Human Visual System
- ✓ Light and the Electromagnetic Spectrum
- ✓ Image Sensing and Acquisition
- ✓ Image Representation
- ✓ Sampling, Quantisation, and Resolution

Digital Image

A **digital image** is a representation of a two-dimensional image as a finite set of digital values, called picture elements or pixels.

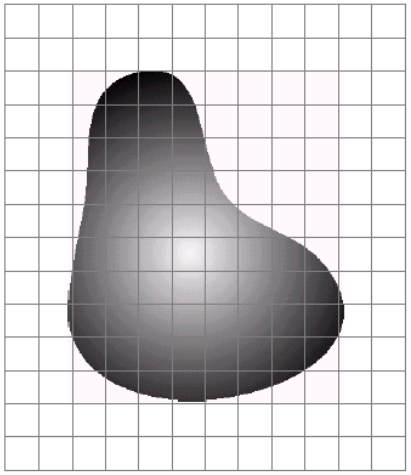


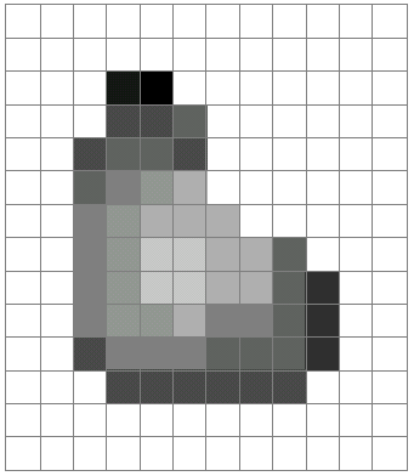
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

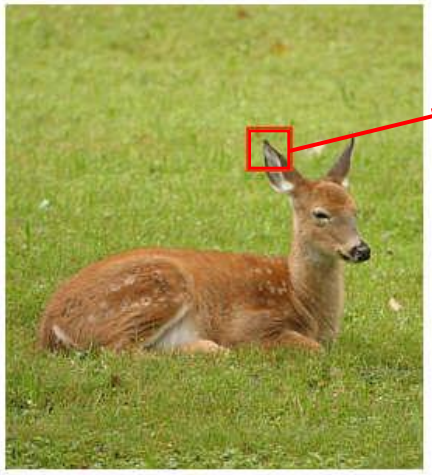
Digital Image

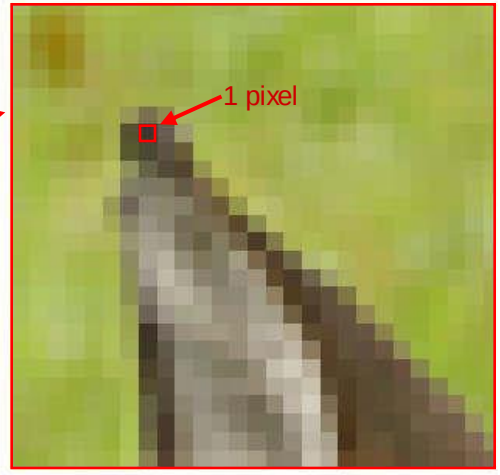
Pixel values typically represent gray levels, colours, heights, opacities, etc.

Digitization implies that a digital image is an *approximation* of a real scene.









Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Digital Image

Common image formats include:

1 sample per point (B&W or Grayscale)

3 samples per point (Red, Green, and Blue)





Digital Image Processing

Digital image processing focuses on two major tasks.

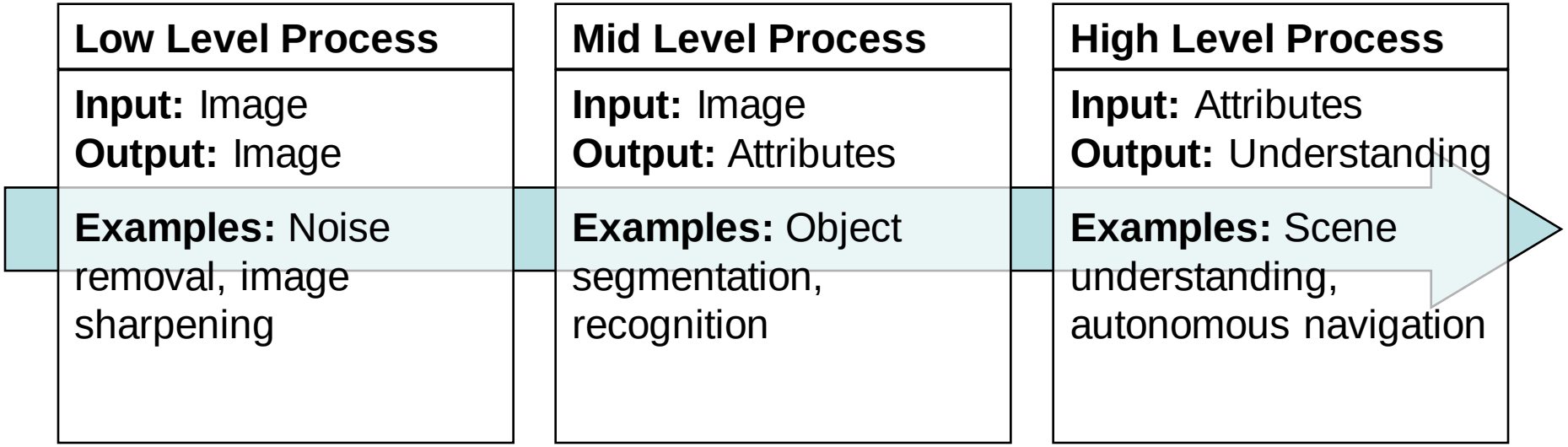
Improvement of pictorial information for human interpretation.

Processing of image data for storage, transmission and representation for autonomous machine perception.

Some argument about where image processing ends and fields such as image analysis and computer vision start.

Digital Image Processing

The continuum from image processing to computer vision can be broken up into low-, mid- and high-level processes.



History of Digital Image Processing

Early 1920s: One of the first applications of digital imaging was in the news-paper industry.

The Bartlane cable picture transmission system.



Early digital image used 5 gray levels

Images were transferred over submarine cable between London and New York.

Pictures were coded for cable transfer and reconstructed at the receiving end on a telegraph printer.

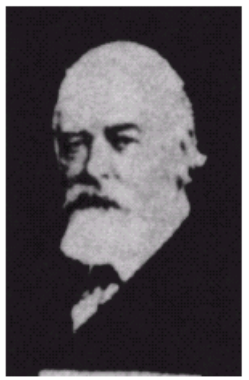
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

History of Digital Image Processing

Mid to late 1920s: Improvements to the Bartlane system resulted in higher quality images.

New reproduction processes based on photographic techniques.

Increased number of tones in reproduced images.



Improved digital image



Early 15 tone digital image

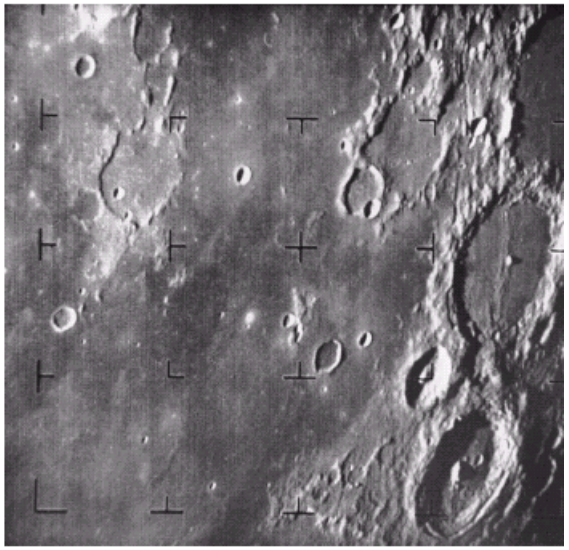
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

History of Digital Image Processing

1960s: Improvements in computing technology and the onset of the space race led to a surge of work in digital image processing.

1964: Computers used to improve the quality of images of the moon taken by the *Ranger 7* probe.

Such techniques were used in other space missions including the Apollo landings.



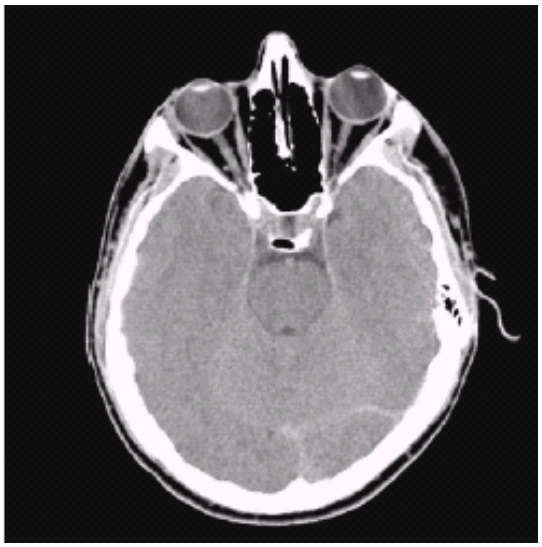
A picture of the moon taken by the Ranger 7 probe minutes before landing

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

History of Digital Image Processing

1970s: Digital image processing begins to be used in medical applications.

1979: Sir Godfrey N. Hounsfield & Prof. Allan M. Cormack share the Nobel Prize in medicine for the invention of tomography, the technology behind Computerised Axial Tomography (CAT) scans.



Typical head slice CAT image

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

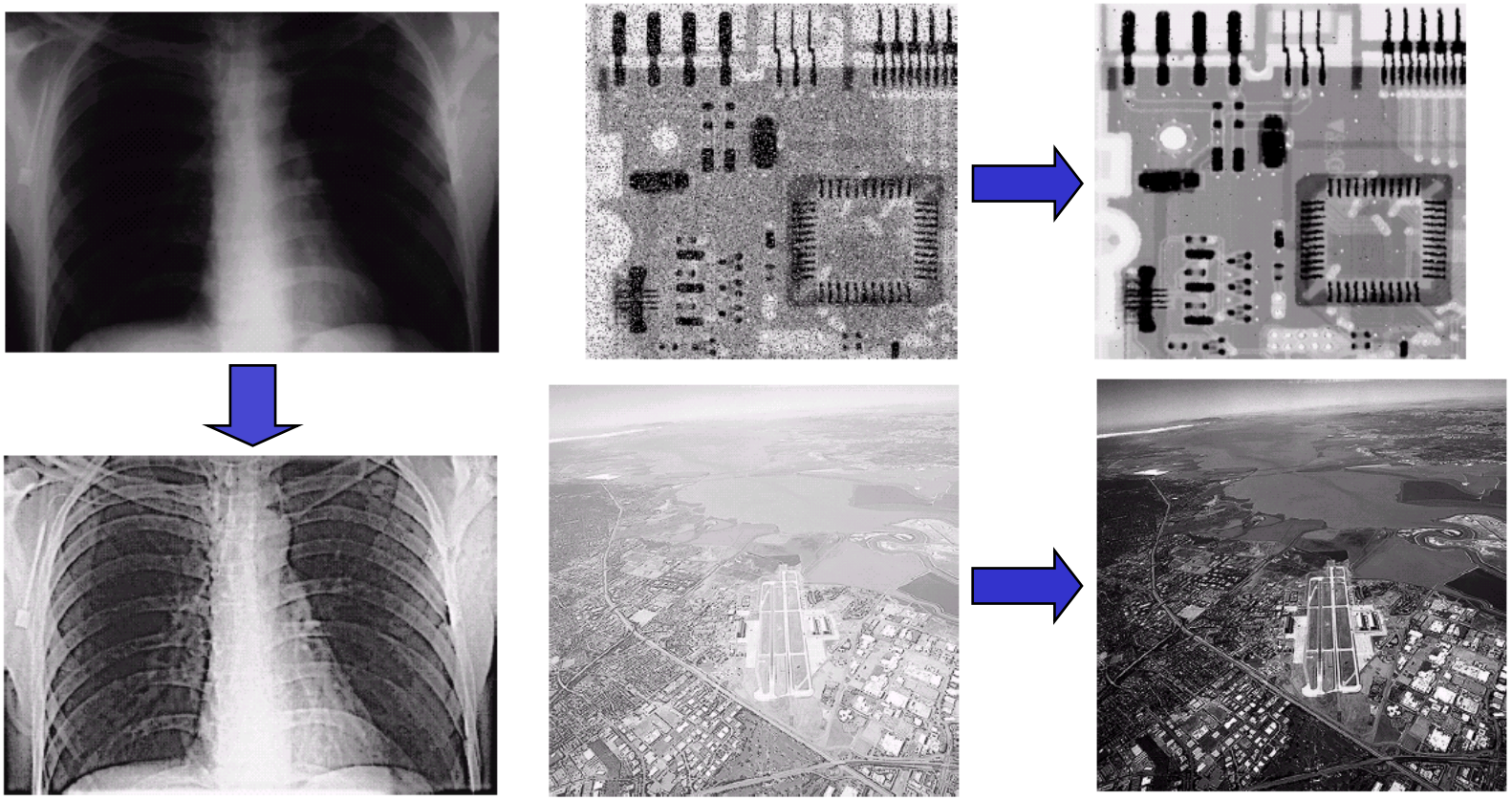
History of Digital Image Processing

1980s - Today: The use of digital image processing techniques has exploded and they are now used for all kinds of tasks in all kinds of areas.

- ✓ Image enhancement/restoration
- ✓ Artistic effects
- ✓ Medical visualisation
- ✓ Industrial inspection
- ✓ Law enforcement
- ✓ Human computer interfaces

Examples: Image Enhancement

One of the most common uses of DIP techniques: improve quality, remove noise, etc.



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Examples: Artistic Effects

Artistic effects are used to make images more visually appealing, to add special effects and to make composite images.





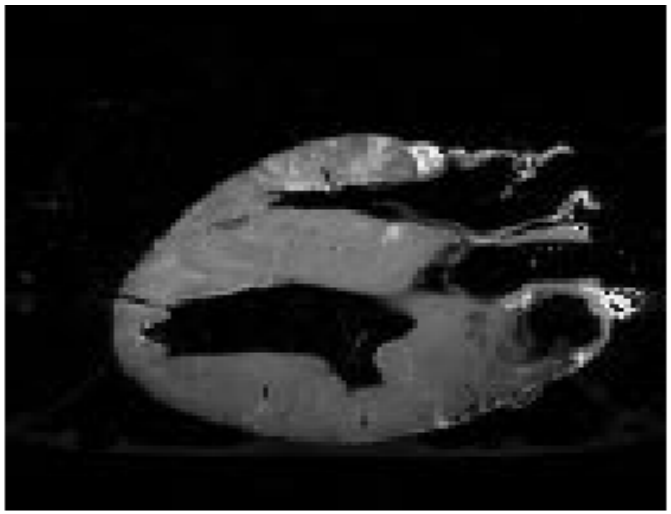




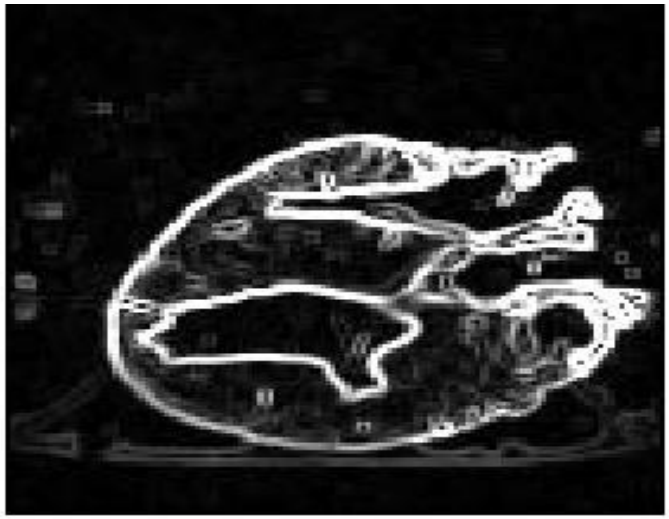
Examples: Medicine

Take slice from MRI scan of a dog heart and find boundaries between types of tissue.

Image with gray levels representing tissue density.



Original MRI Image of a Dog Heart



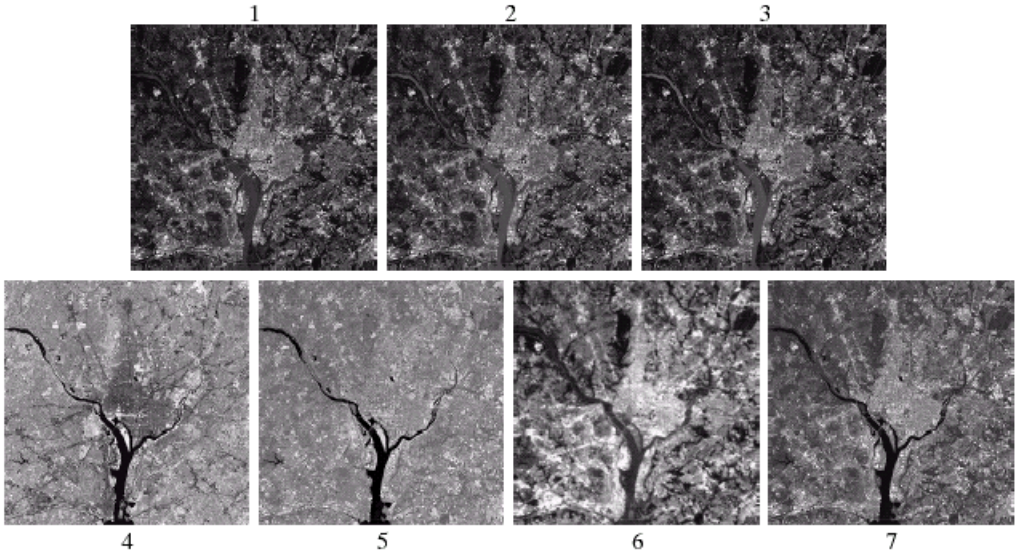
Edge Detection Image

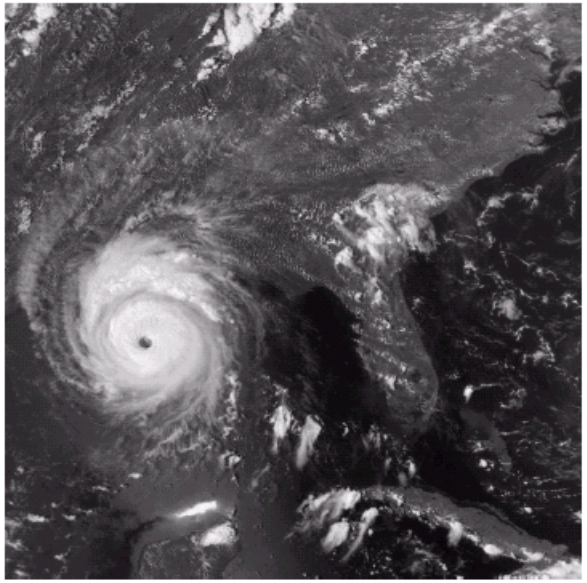
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Examples: GIS

Geographic Information Systems

Digital image processing techniques are used extensively to manipulate satellite imagery.





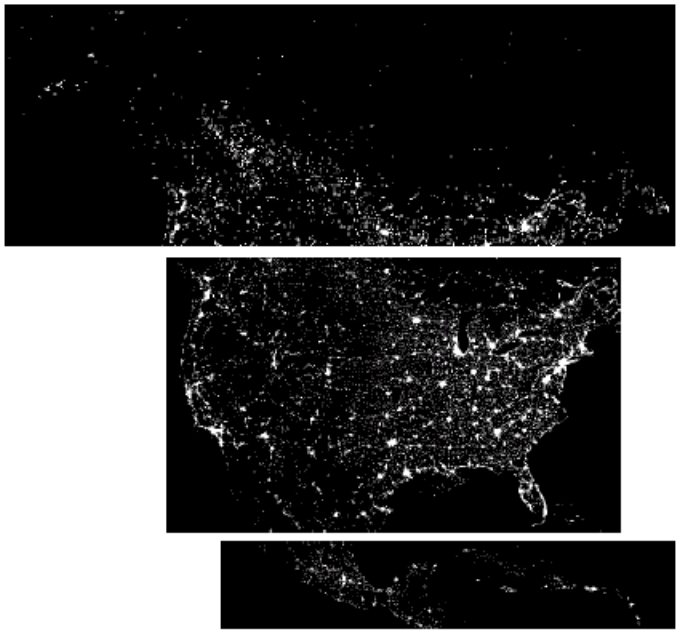
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Examples: GIS

Night-Time Lights of the World data set.

Global inventory of human settlement.

Not hard to imagine the kind of analysis that might be done using this data.







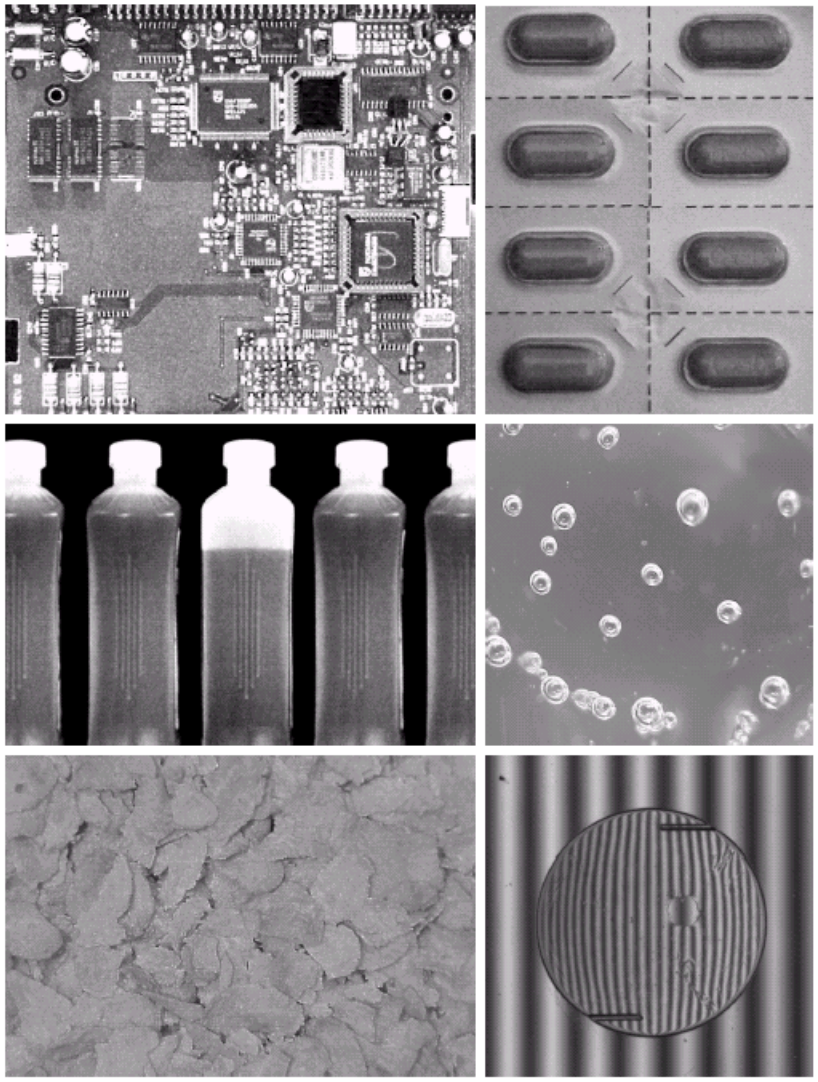
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Examples: Industrial Inspection

Human operators are expensive, slow and unreliable.

Make machines do the job instead.

Industrial vision systems are used in all kinds of industries.



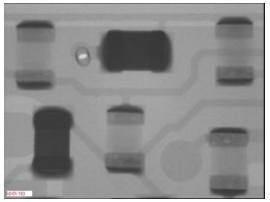
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

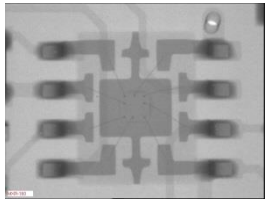
Examples: PCB Inspection

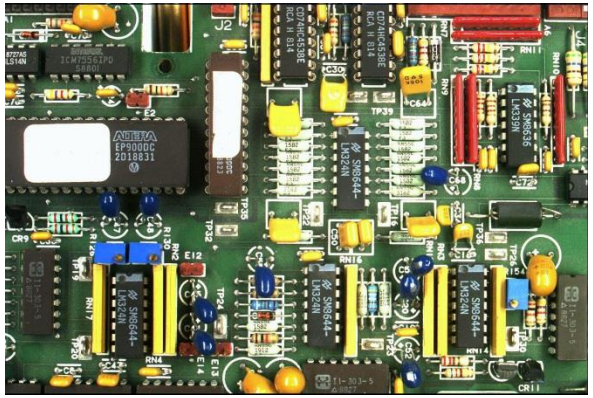
Printed Circuit Board (PCB) inspection

- Machine inspection is used to determine that all components are present and that all solder joints are acceptable.
- Both conventional imaging and x-ray imaging are used.









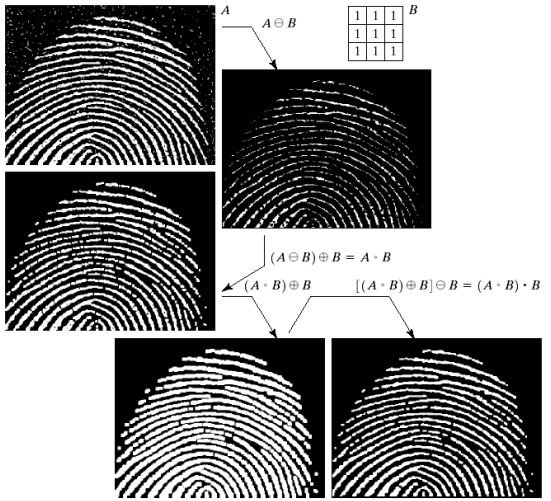


Examples: Law Enforcement

Image processing techniques are used extensively by law enforcers.

- Number plate recognition for speed cameras or automated toll systems
- Fingerprint recognition
- Enhancement of CCTV images



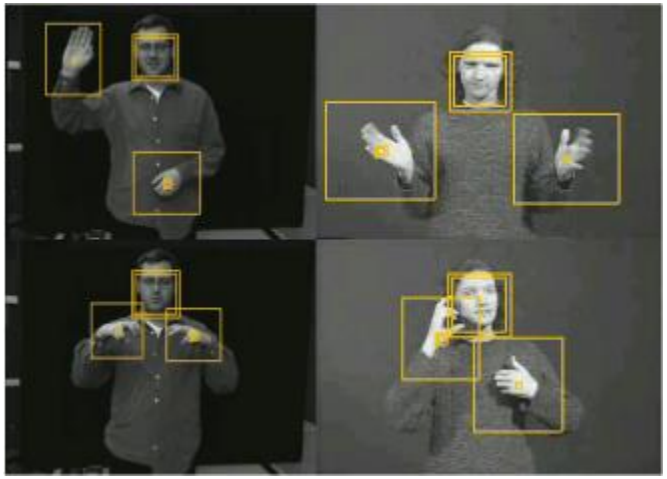


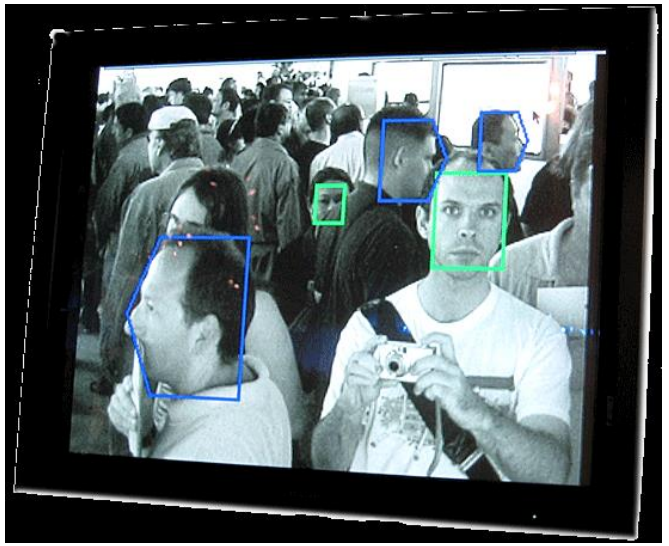
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Examples: HCI

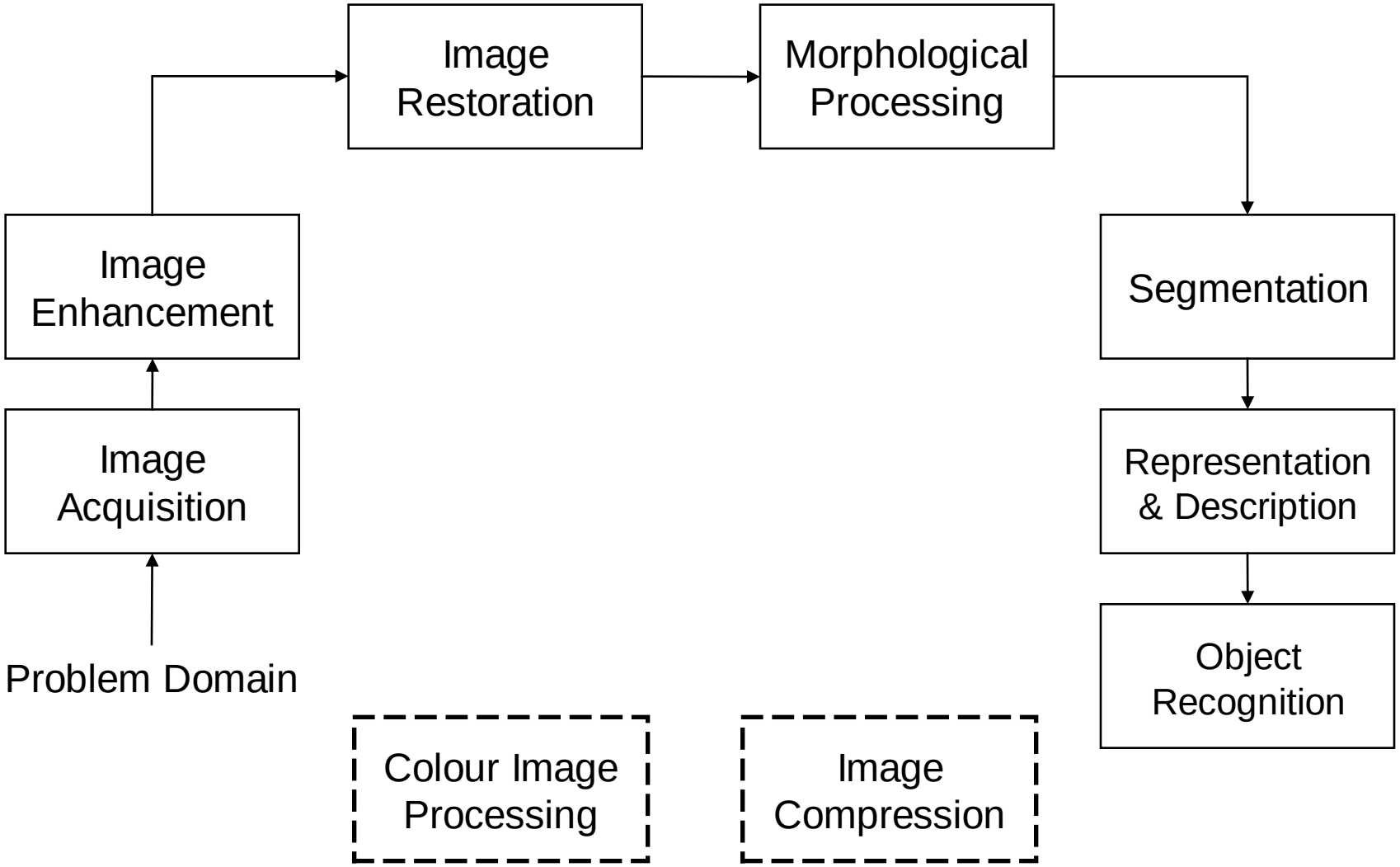
Try to make human computer interfaces more natural

- Face recognition
- Gesture recognition
- Handwriting recognition
- Optical character recognition (OCR)
- …

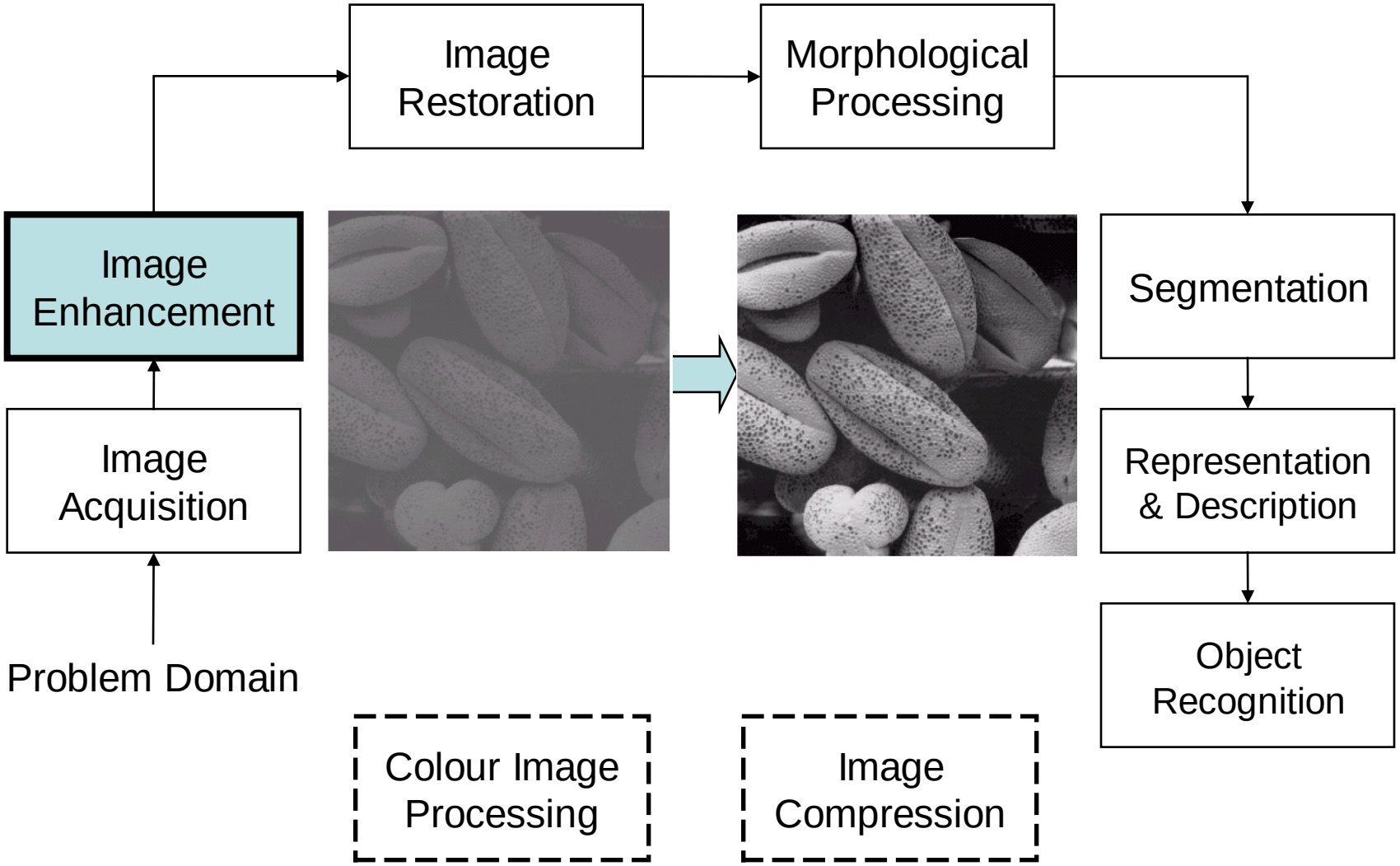




Phases of Digital Image Processing

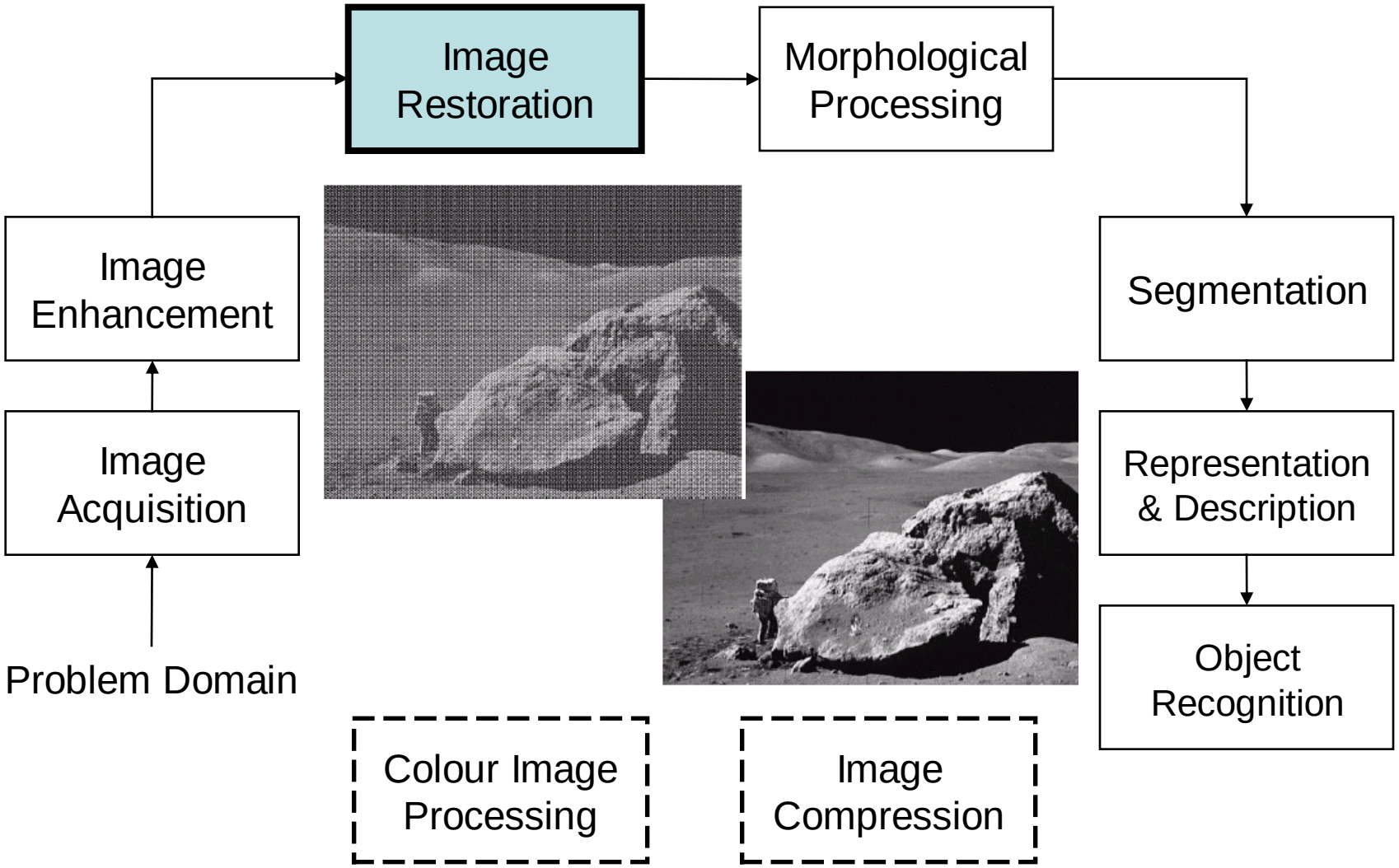


Phases of Digital Image Processing : Image Enhancement



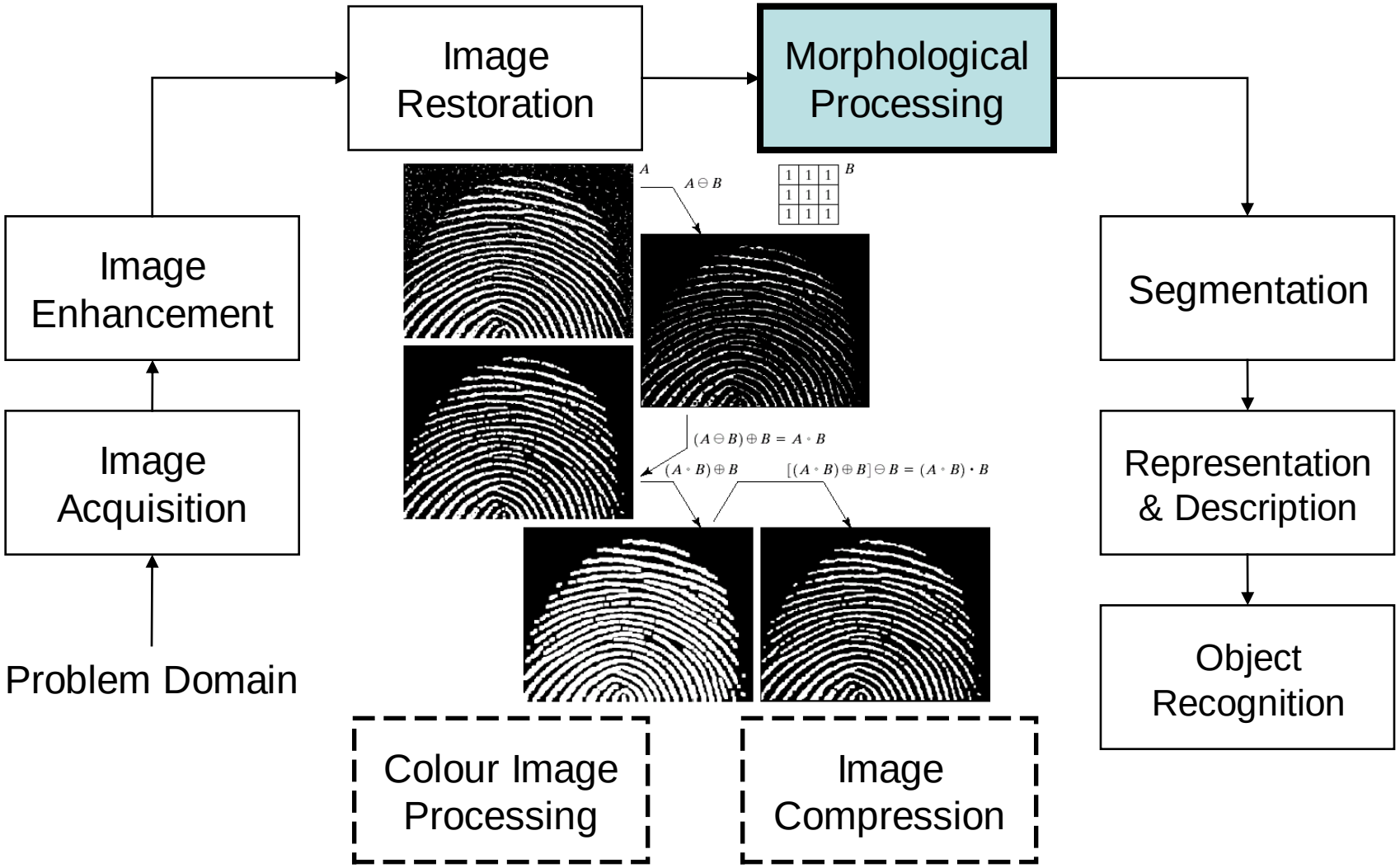
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Phases of Digital Image Processing : Image Restoration



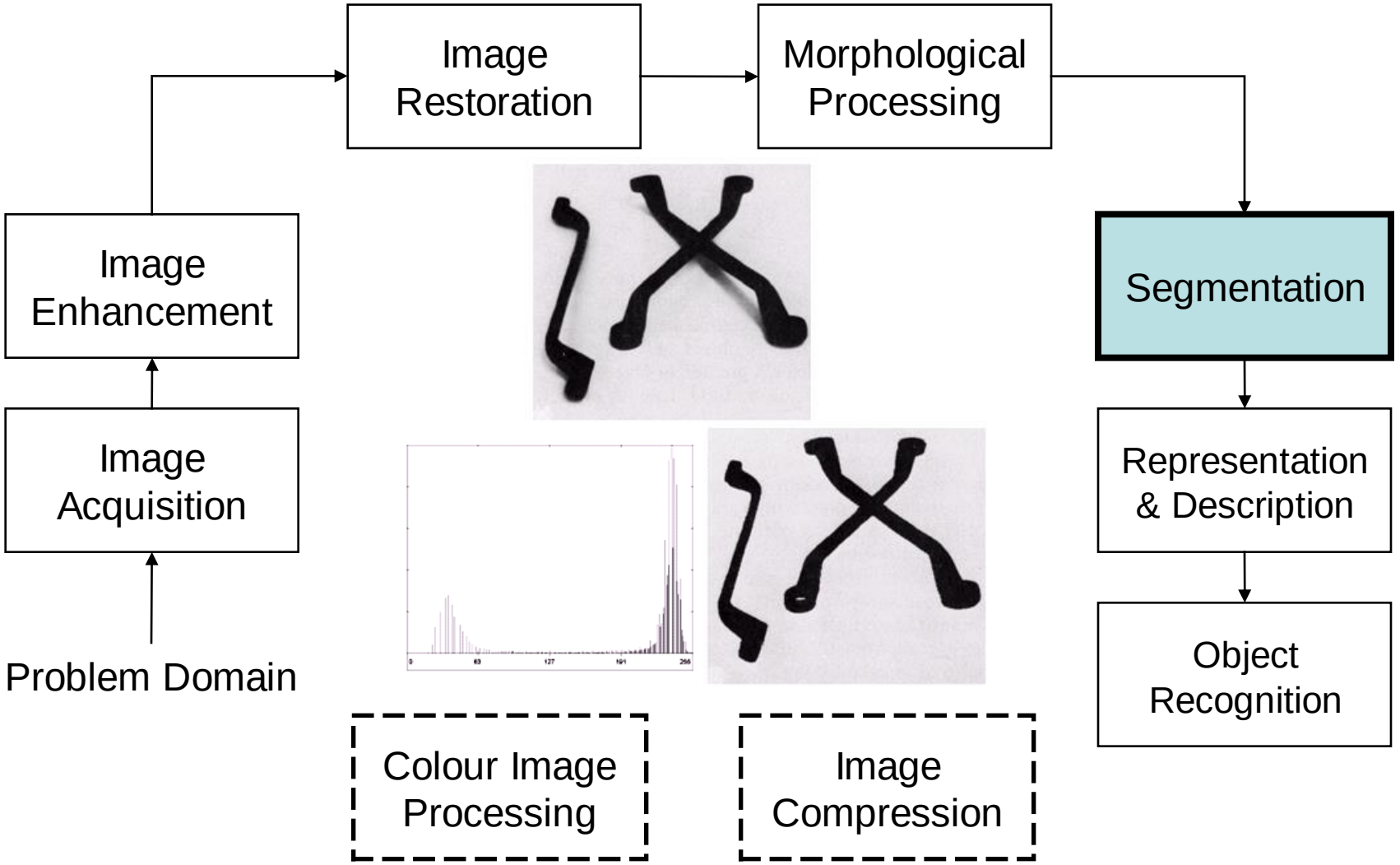
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Phases of Digital Image Processing : Morphological Processing



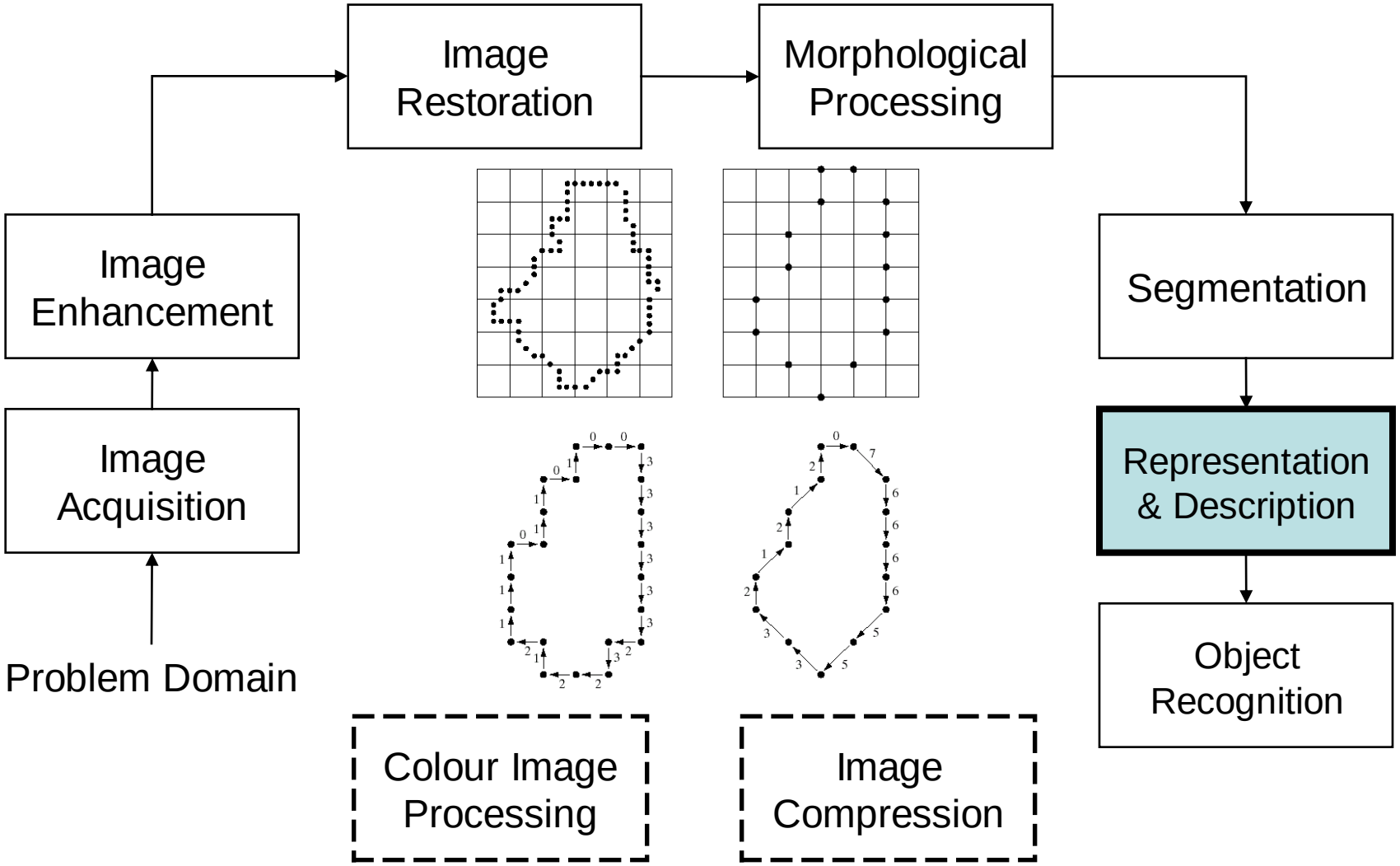
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Phases of Digital Image Processing : Segmentation



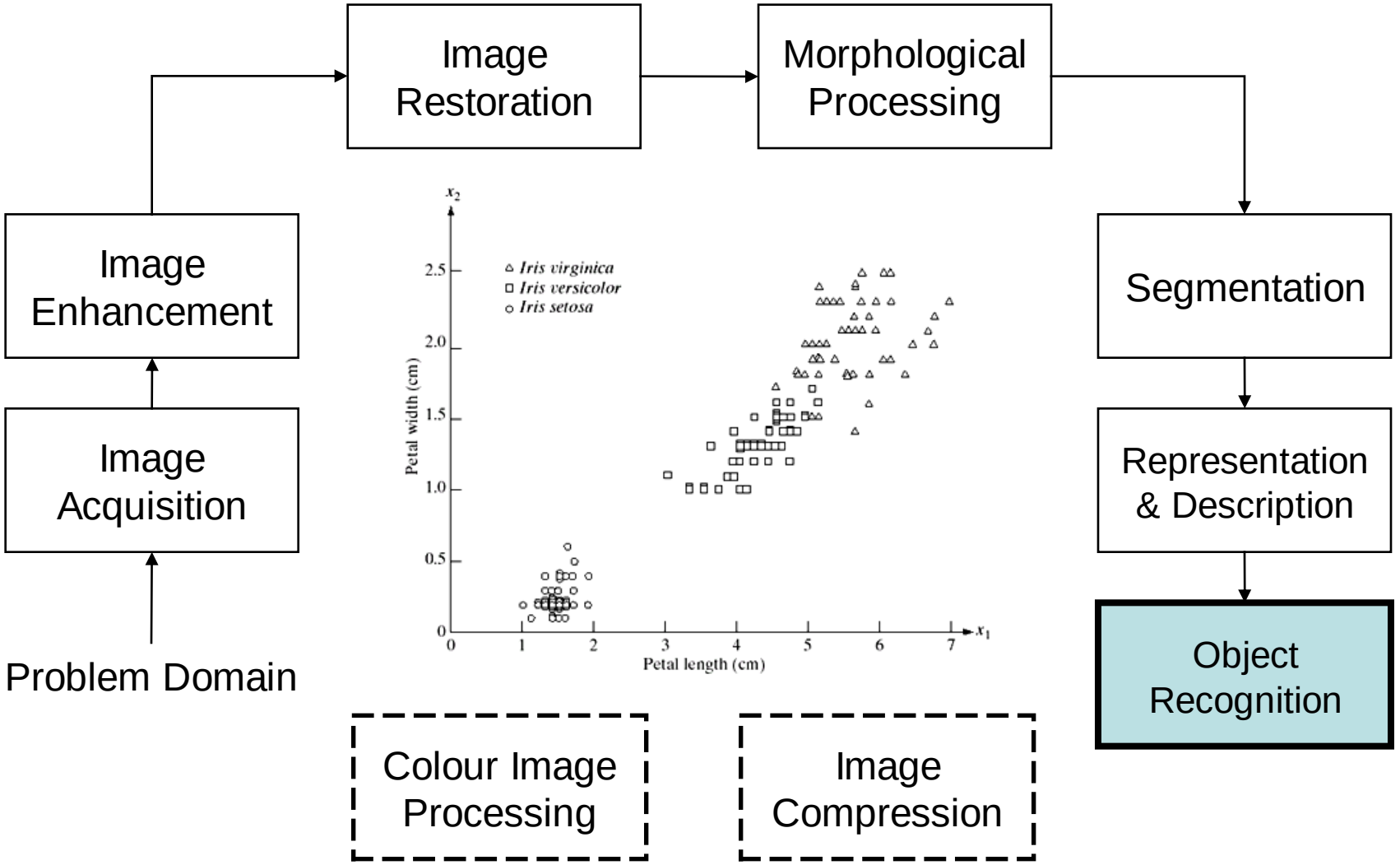
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Phases of Digital Image Processing : Representation & Description



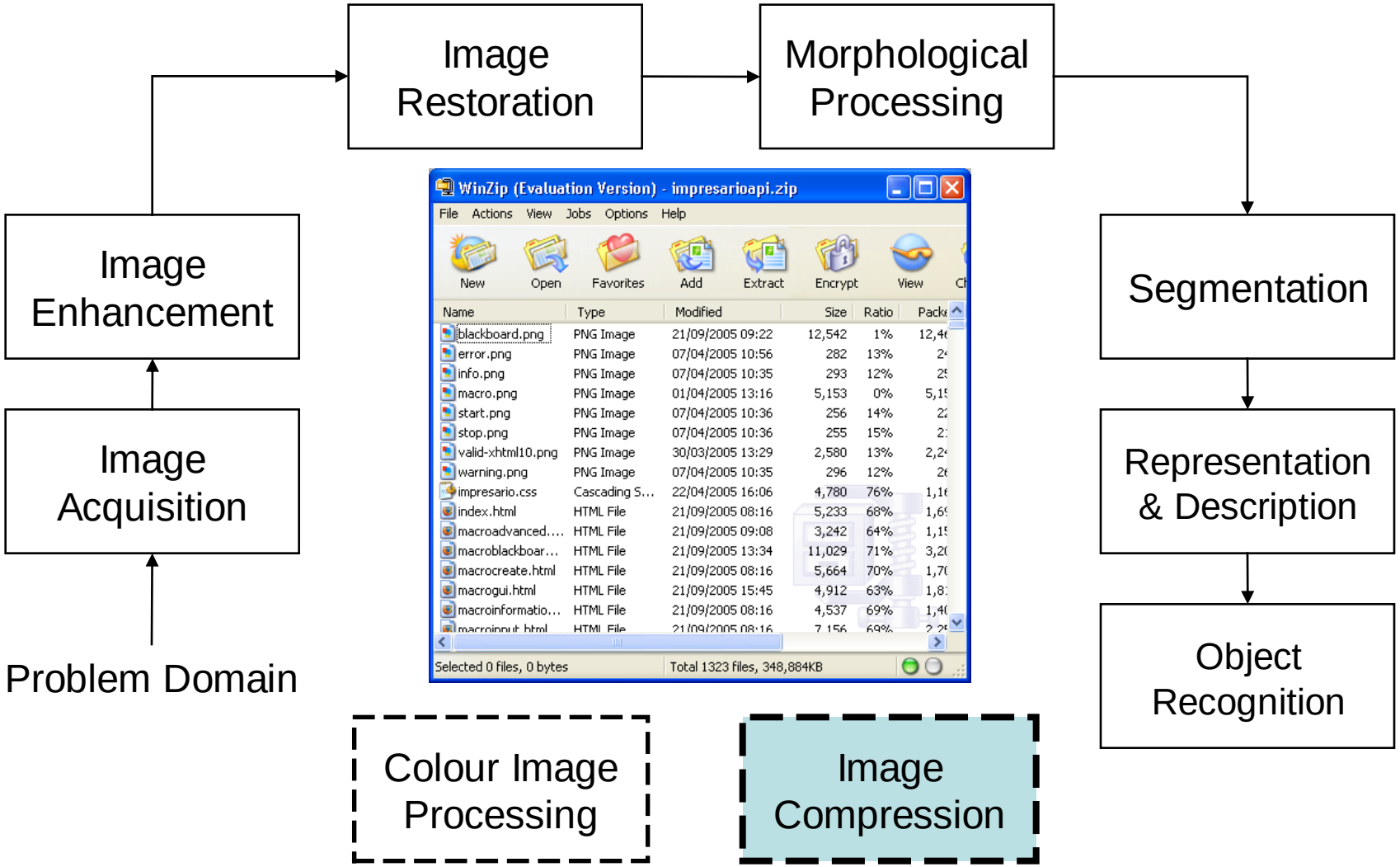
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Phases of Digital Image Processing : Object Recognition

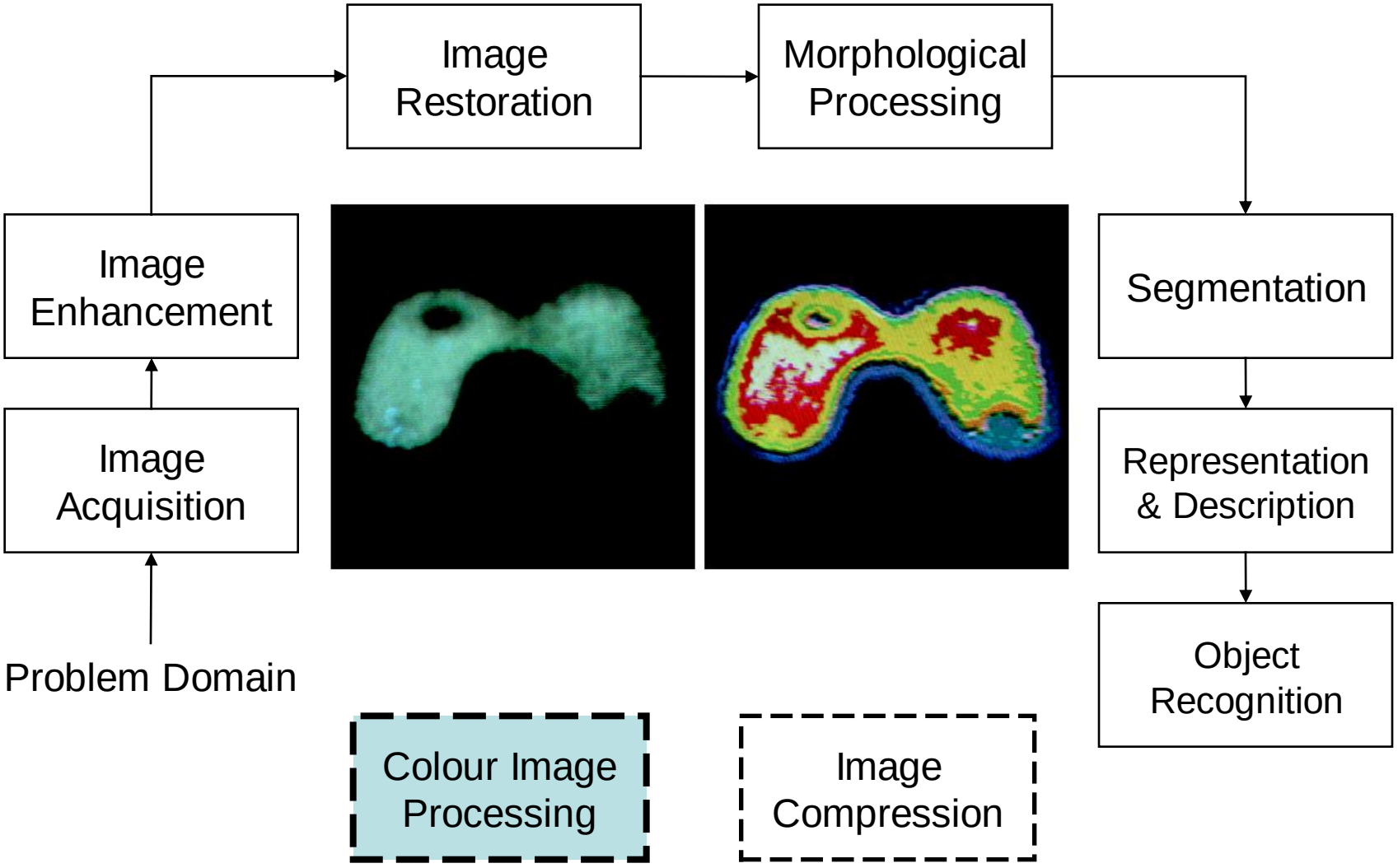


Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Phases of Digital Image Processing : Image Compression



Phases of Digital Image Processing : Colour Image Processing



Human Visual System

The best vision model we have!

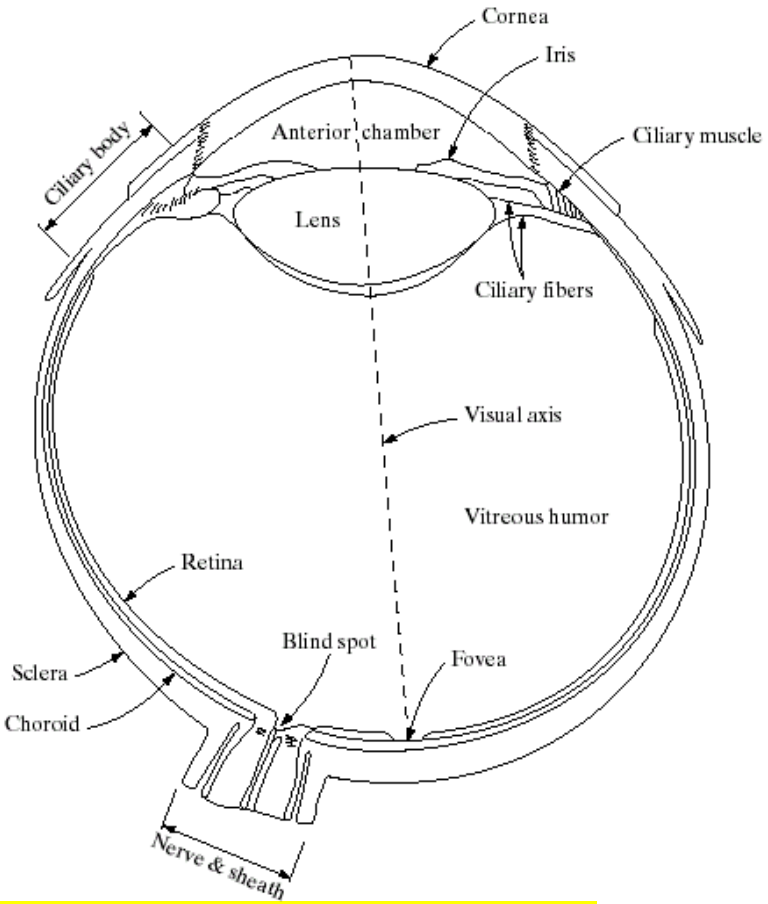
Knowledge of how images form in the eye can help us with processing digital images.

We will take just a rapid tour of the human visual system.

Structure of The Human Eye

The lens focuses light from objects onto the retina.

The retina is covered with light receptors called *cones* (6-7 million) and *rods* (75-150 million).



Cones are concentrated around the fovea and are very sensitive to colour.

Rods are more spread out and are sensitive to low levels of illumination.

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Structure of The Human Eye

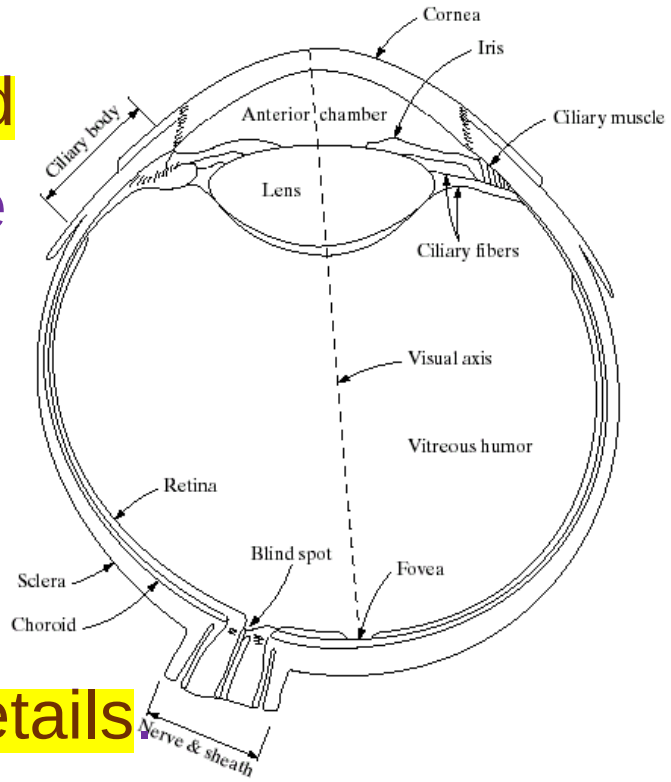
Cornea and sclera keep everything in!

The choroid contains all of the blood vessels that serve as nutrition to the eye.

The iris contains the pigment that gives our eyes their colour

Rods for general vision cones for details

Blind spot is where the optic nerve connects.

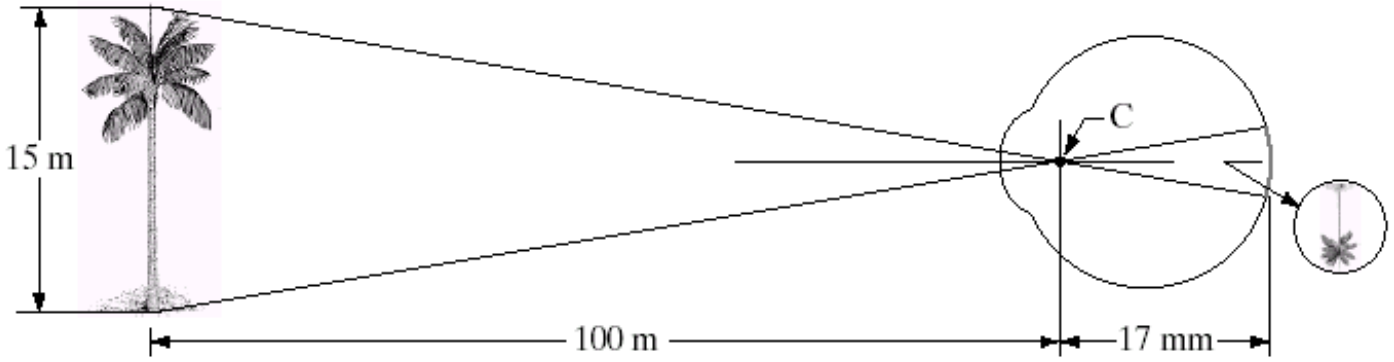


Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Image Formation in the Eye

Muscles within the eye can be used to change the shape of the lens allowing us focus on objects that are near or far away.

An image is focused onto the retina causing rods and cones to become excited which ultimately send signals to the brain.



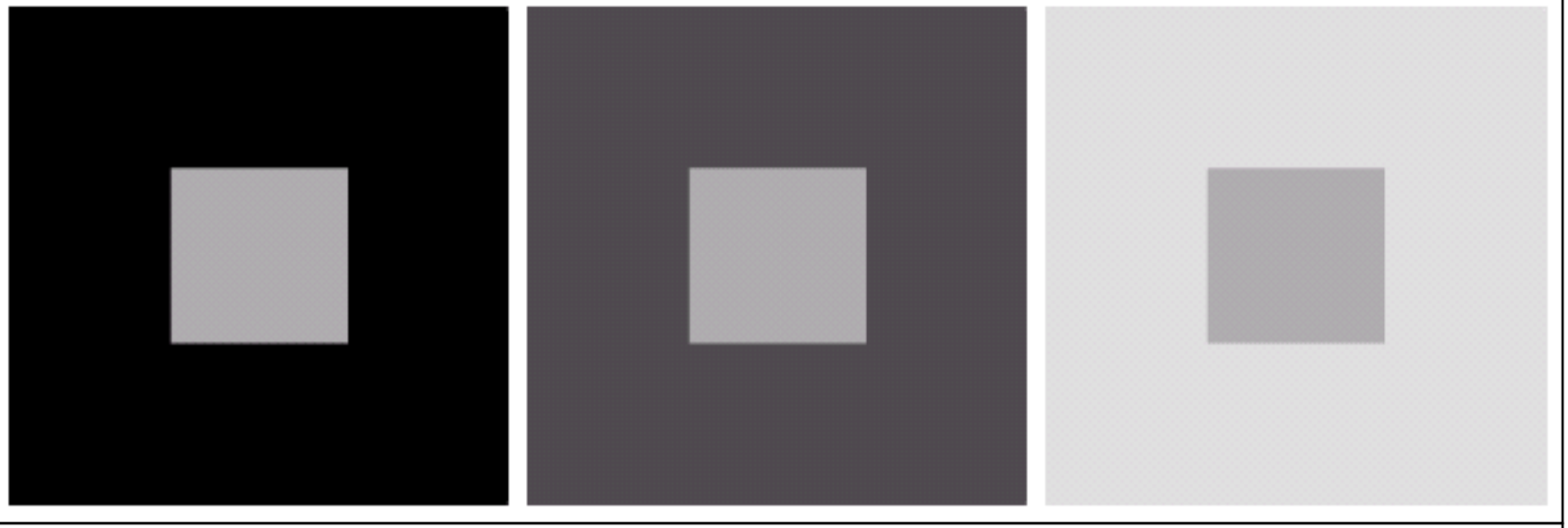
Brightness Adaptation & Discrimination

The human visual system can perceive approximately 10^{10} different light intensity levels.

However, at any one time we can only discriminate between a much smaller number – *brightness adaptation*.

Similarly, the *perceived intensity* of a region is related to the light intensities of the regions surrounding it.

Brightness Adaptation & Discrimination



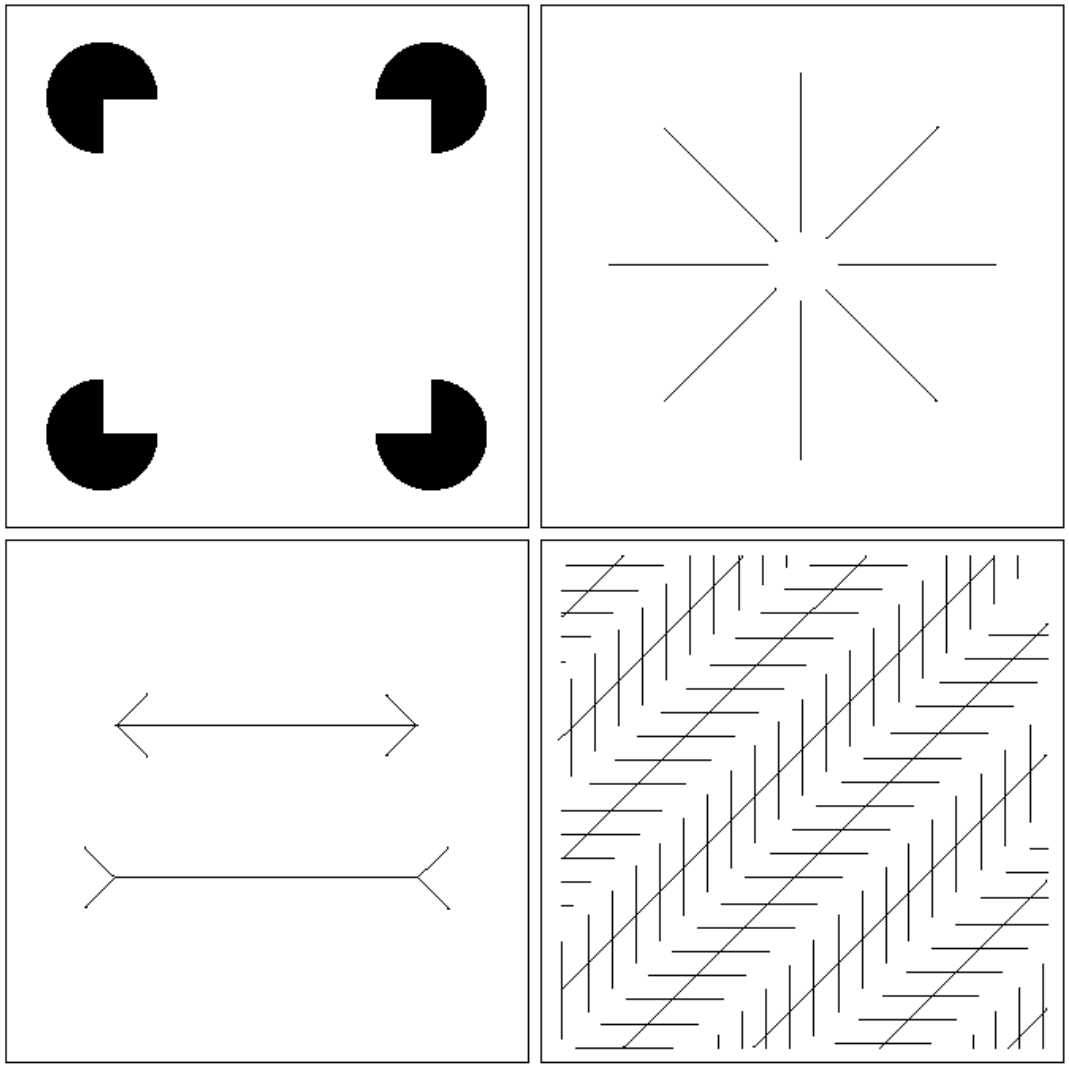
An example of *simultaneous contrast*

All the inner squares have the same intensity, but they appear progressively darker as the background becomes lighter.

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Optical Illusions

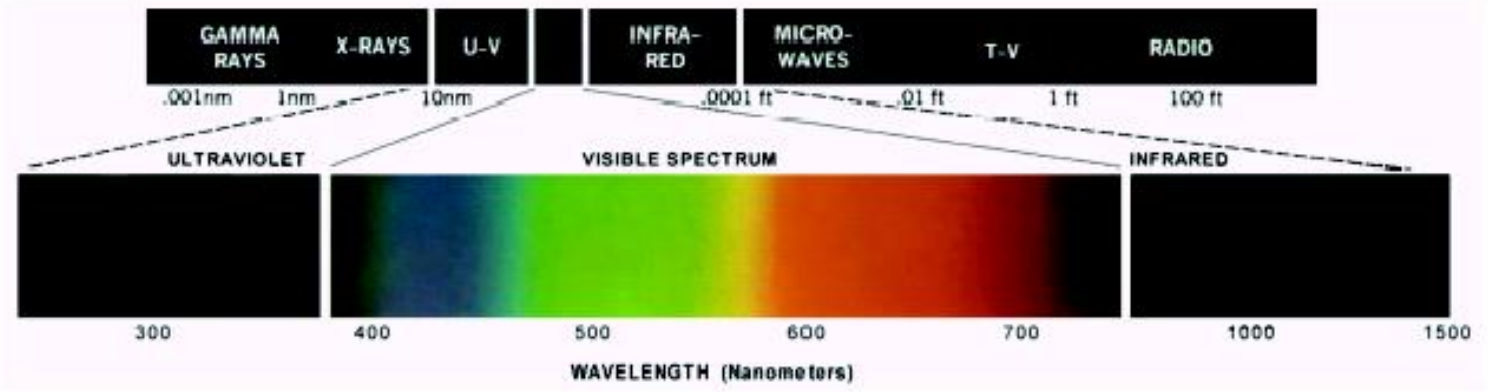
Our visual systems play lots of interesting tricks on us.



Light and Electromagnetic Spectrum

Light is just a particular part of the electromagnetic spectrum that can be sensed by the human eye.

The electromagnetic spectrum is split up according to the wavelengths of different forms of energy.



Light and Electromagnetic Spectrum

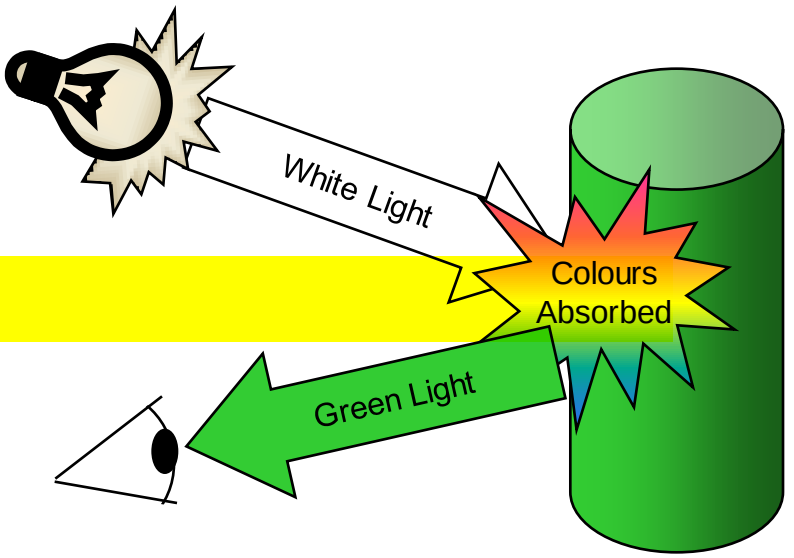
In 1666, Sir Isaac Newton discovered that light passed through a prism splits into a continuous spectrum of colours.

Many image applications use electromagnetic radiation that is far outside the visual spectrum – x-ray images, infra-red images. We generate images of these images by mapping them to the visual spectrum.

Reflected Light

The colours that we perceive are determined by the nature of the light reflected from an object.

For example, if white light is shone onto a green object most wavelengths are absorbed, while green light is reflected from the object.



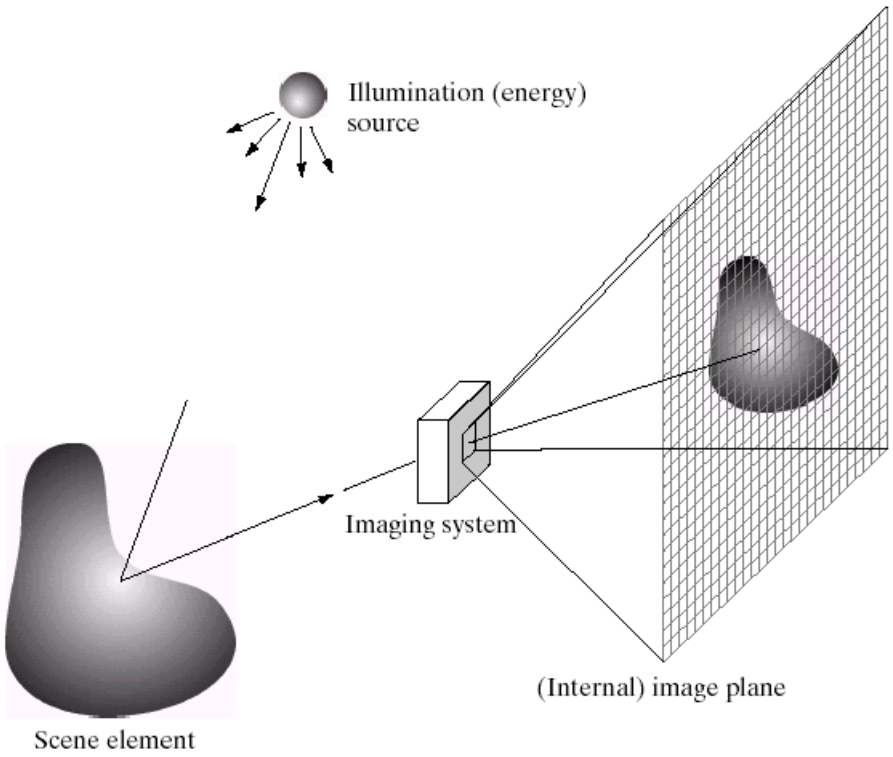
Sampling, Quantisation and Resolution

We consider what is involved in capturing a digital image of a real-world scene.

- ✓ Image Sensing and Acquisition
- ✓ Sampling and Quantisation
- ✓ Representation and Resolution

Image Sensing and Acquisition

Images are typically generated by *illuminating* a *scene* and absorbing the energy reflected by the objects in that scene.



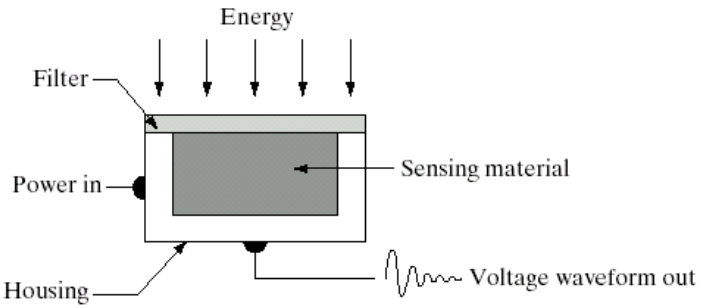
– Typical notions of illumination and scene can be way off:
  - X-rays of a skeleton
  - Ultrasound of an unborn baby
  - Electro-microscopic images of molecules

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Image Sensing and Acquisition

Incoming energy lands on a sensor material responsive to that type of energy and this generates a voltage.

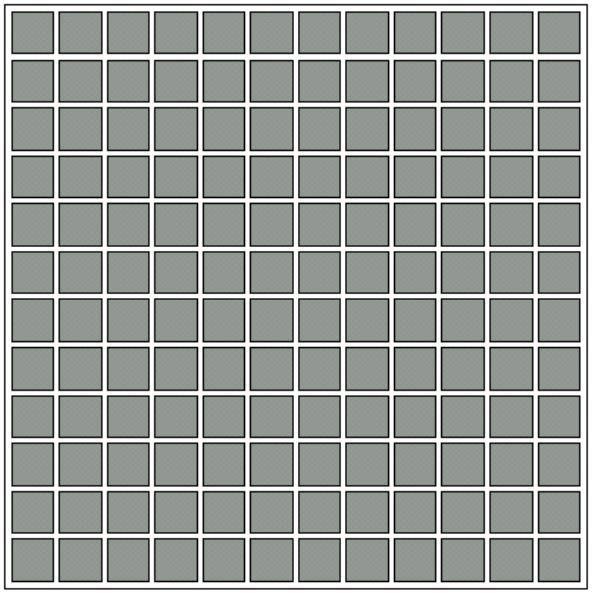
Collections of sensors are arranged to capture images.



Imaging Sensor



Line of Image Sensors



Array of Image Sensors

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Digitization

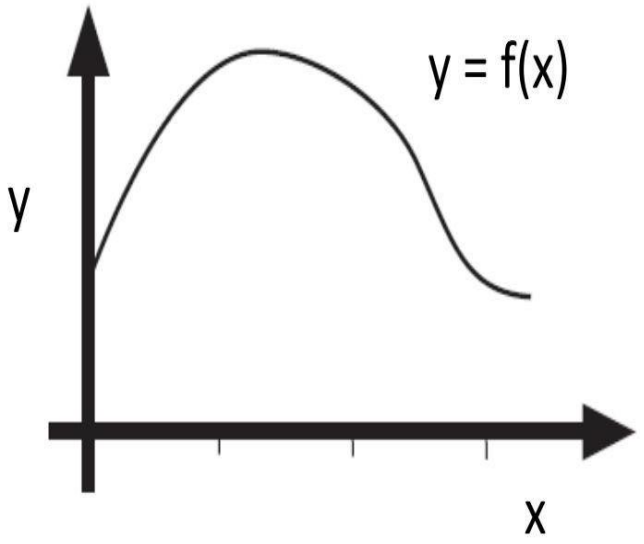


Digitization

- Function

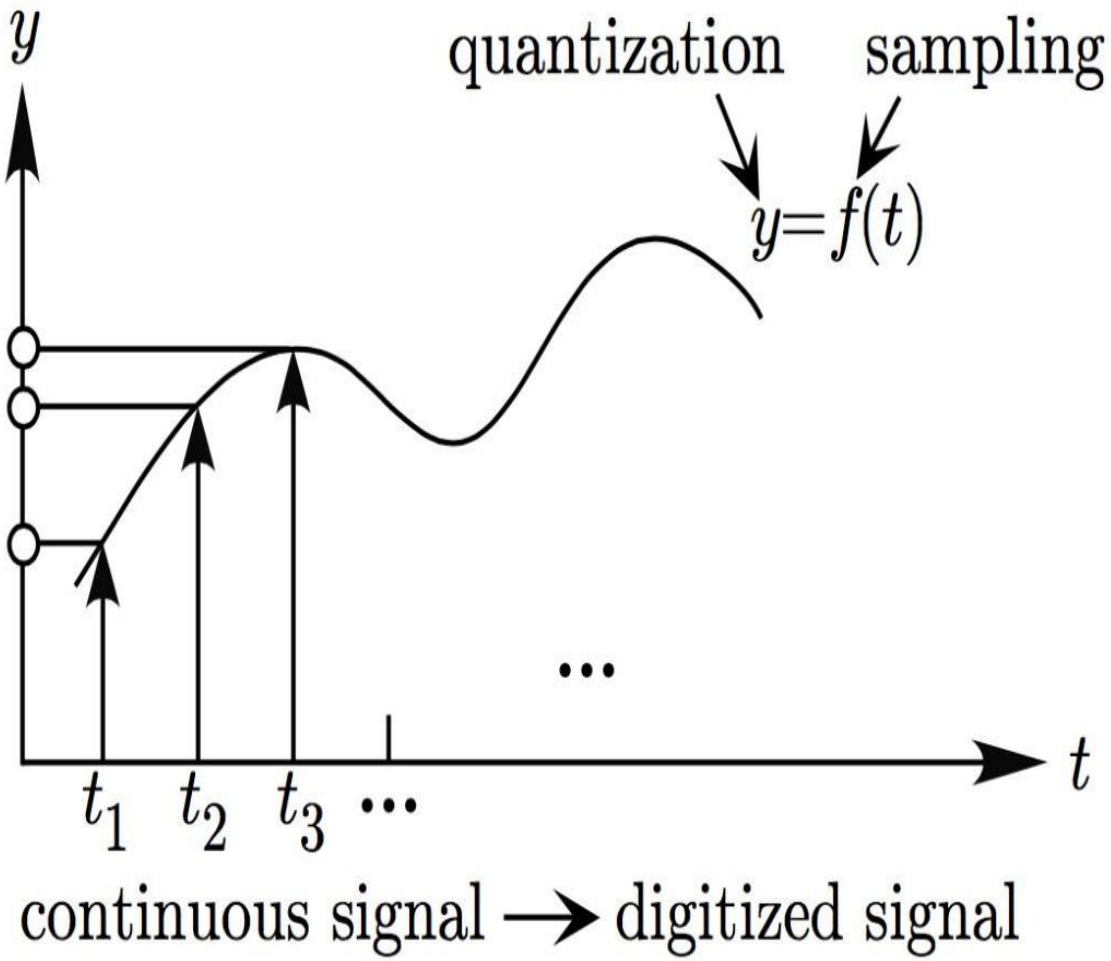
  $y = f(x)$

- Domain of a function
- Range of a function
- Sampling
  - Discretization of domain
- Quantization
  - Discretization of range



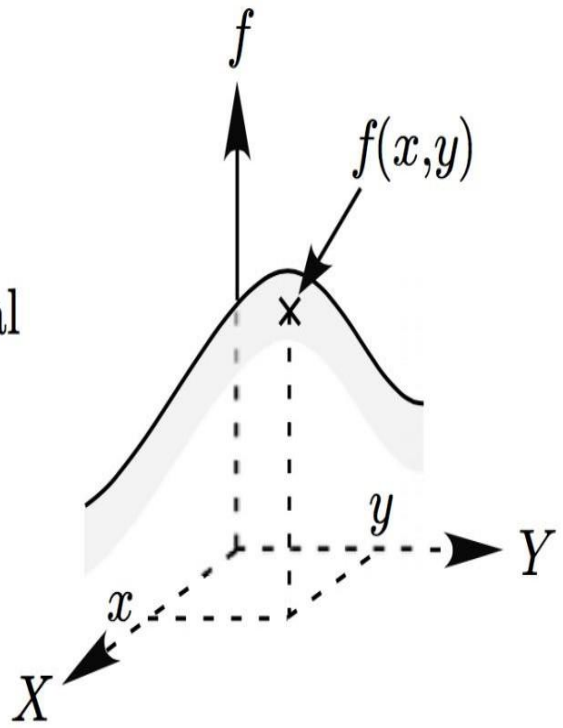
Digitization of a 1D Function

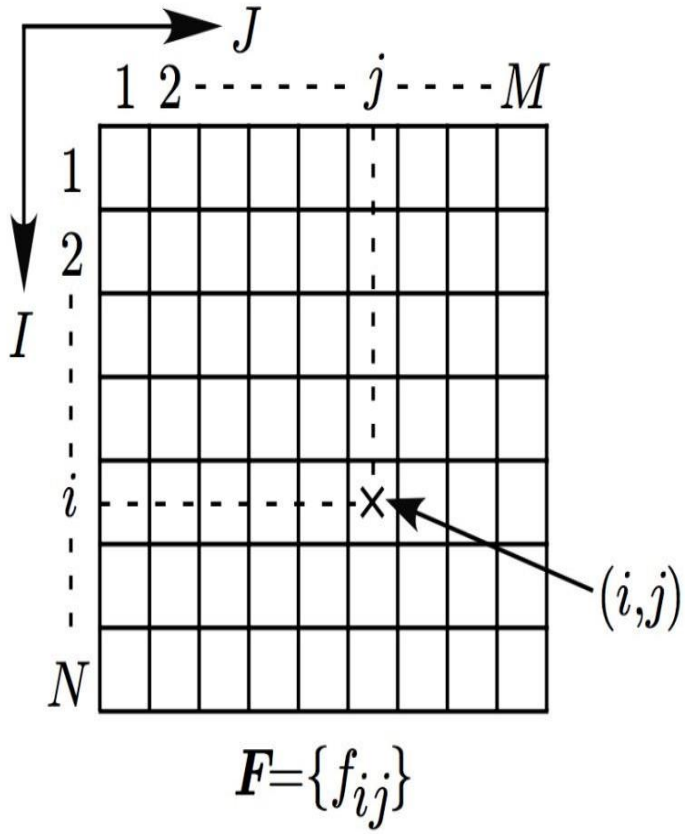
one-dimensional



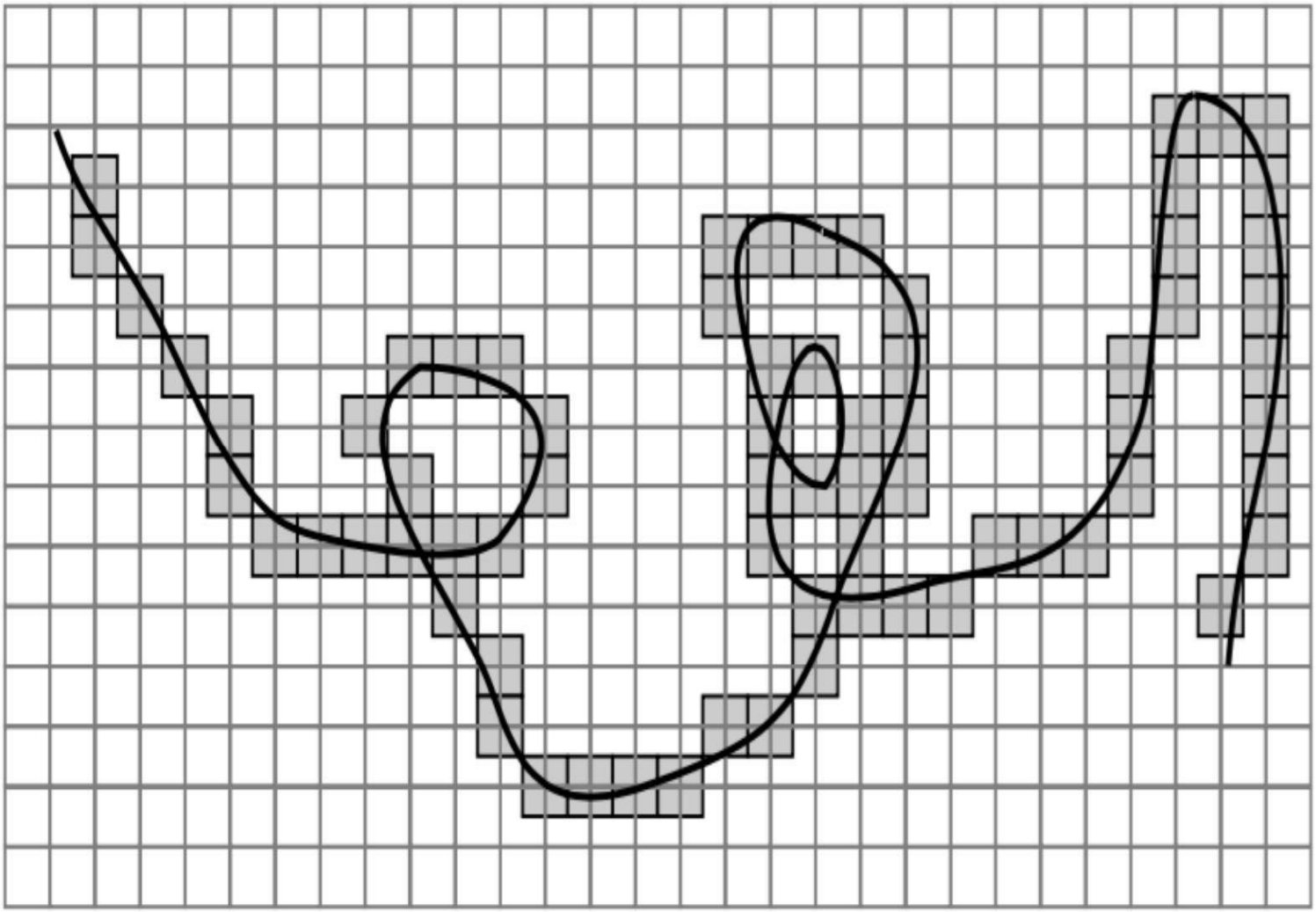
Digitization of a 2D Function

two-dimensional





Digitization of an Arc



Gray Scale Digital Image

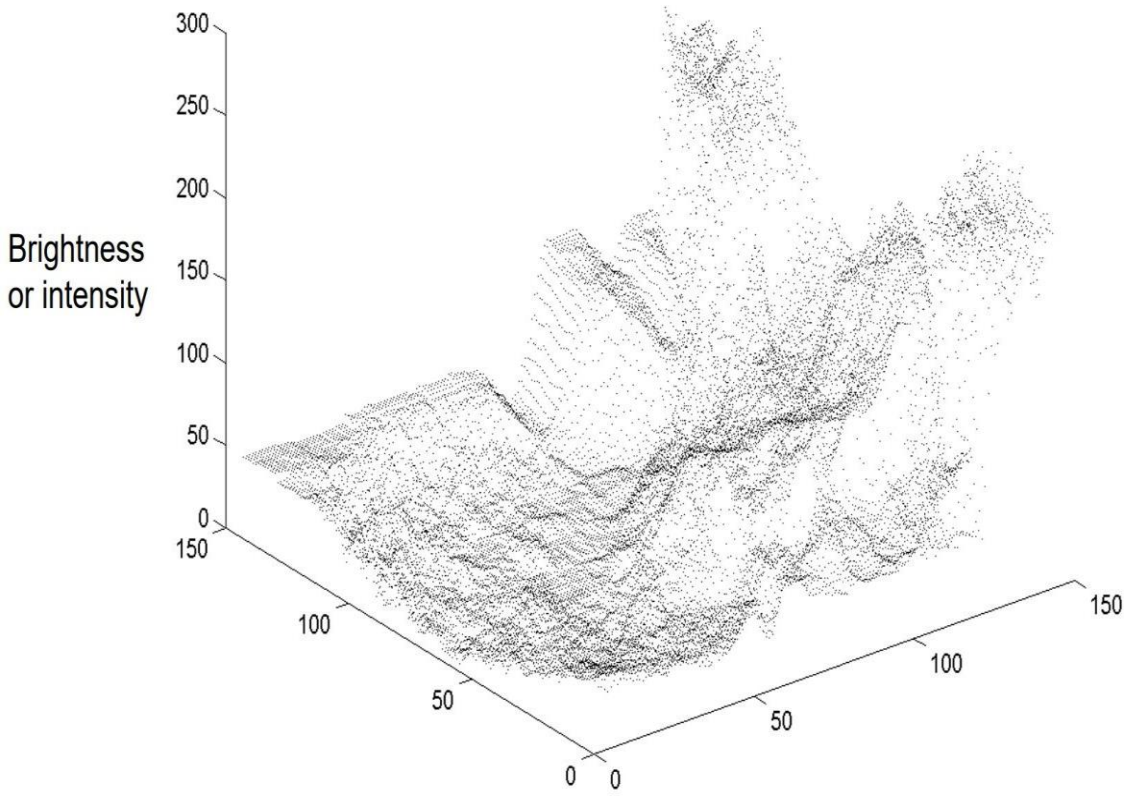
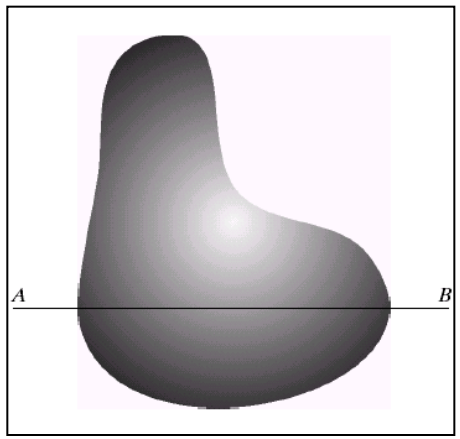


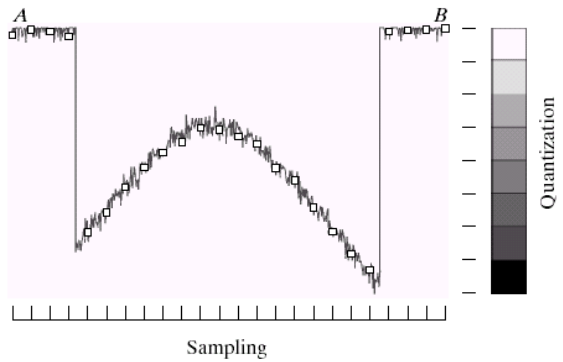
Image Credit Danny Alexander

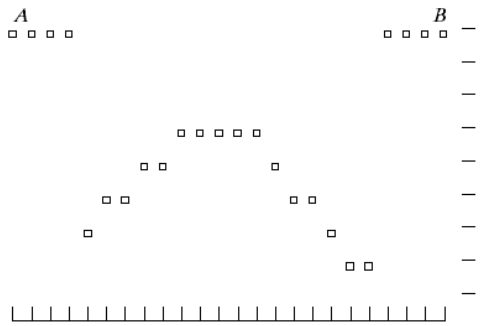
Image Sampling and Quantisation

A digital sensor can only measure a limited number of **samples** at a **discrete** set of energy levels.

Sampling and Quantisation is the process of converting a continuous **analogue** signal into a digital representation of this signal.

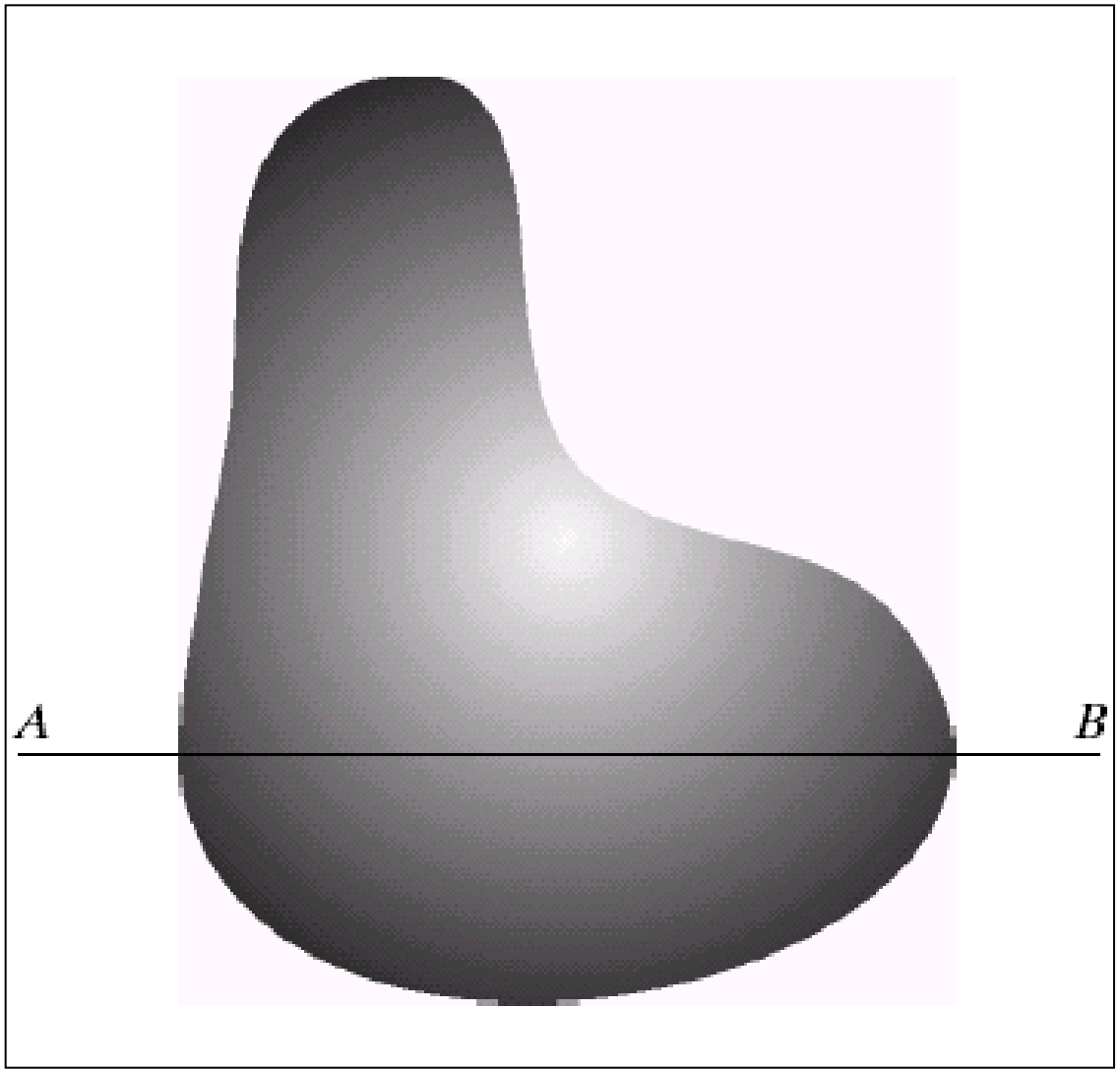






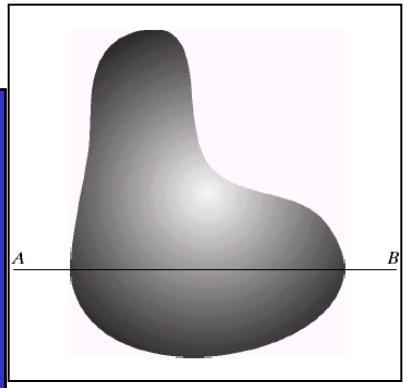
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

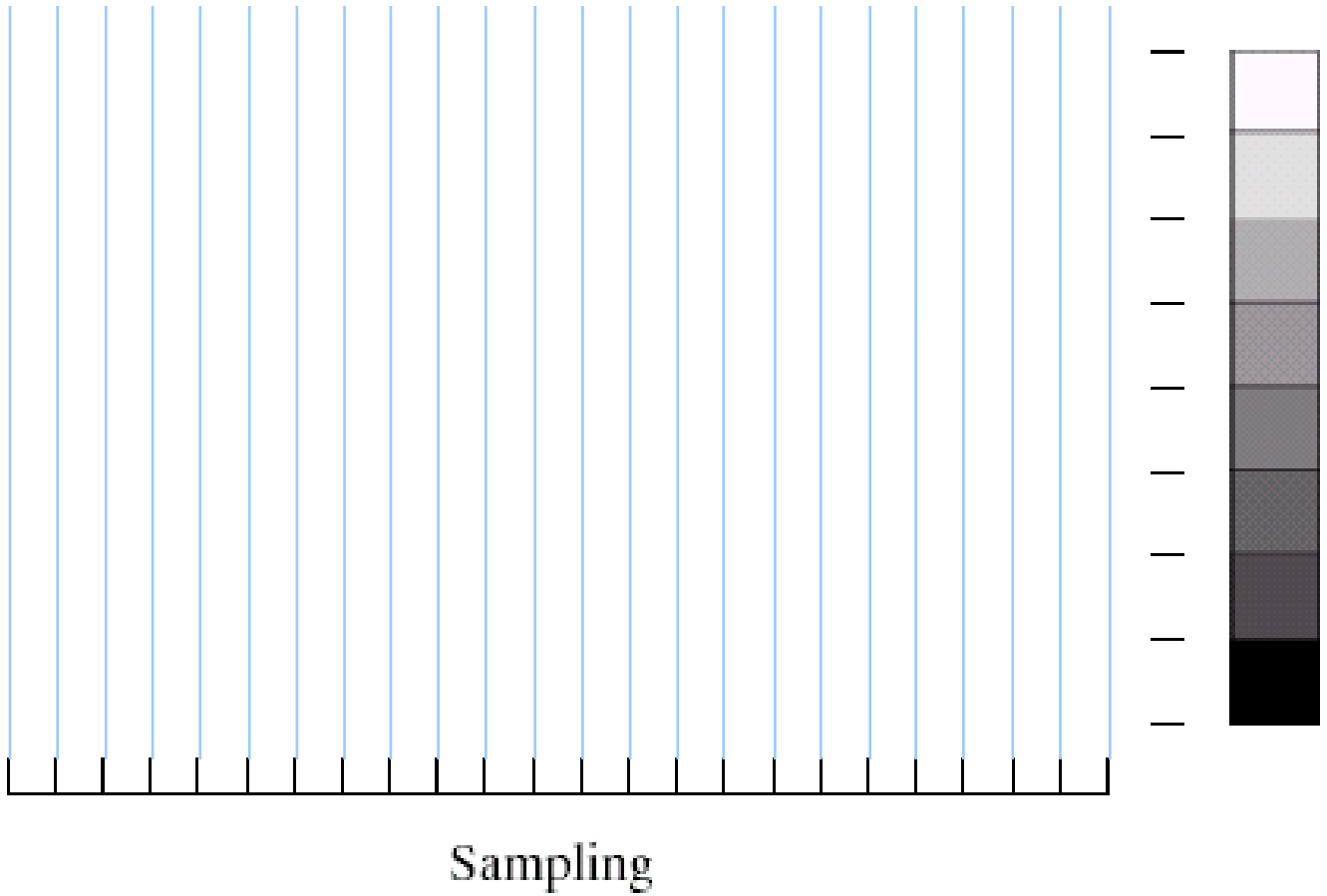
Image Sampling and Quantisation



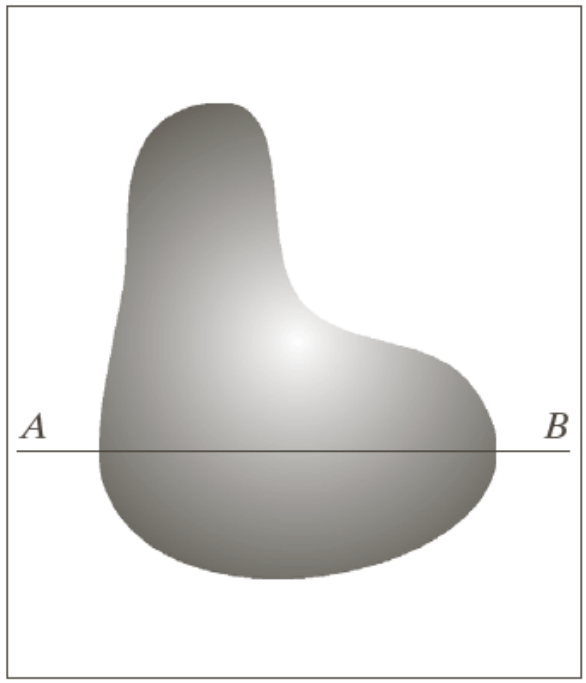
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

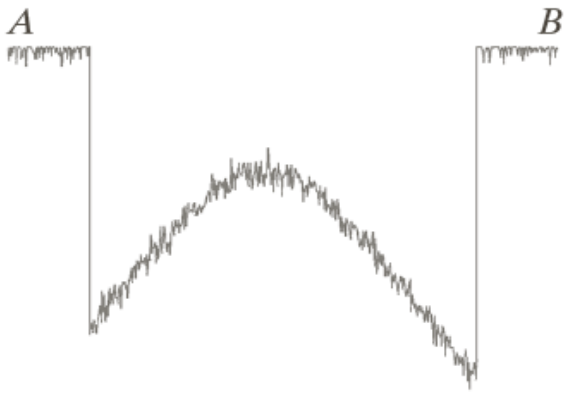
Image Sampling and Quantisation

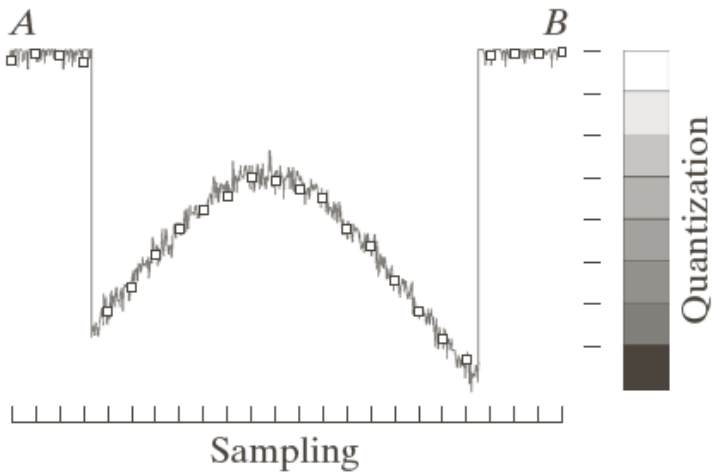




Images taken from Gonzalez & Woods, Digital Image Processing (2002)


A
B


A
B


A
B
Sampling

Quantization

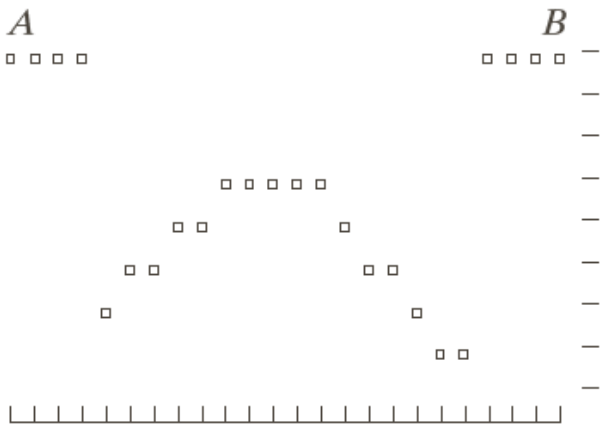
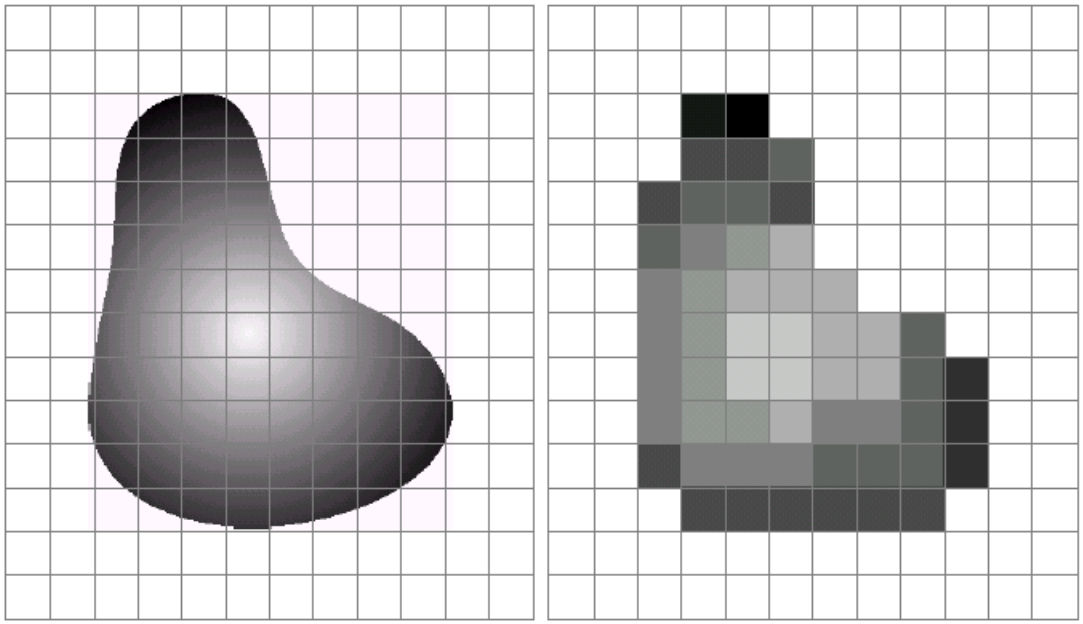

A
B

Image Sampling and Quantisation

Remember that a digital image is always only an **approximation** of a real world scene



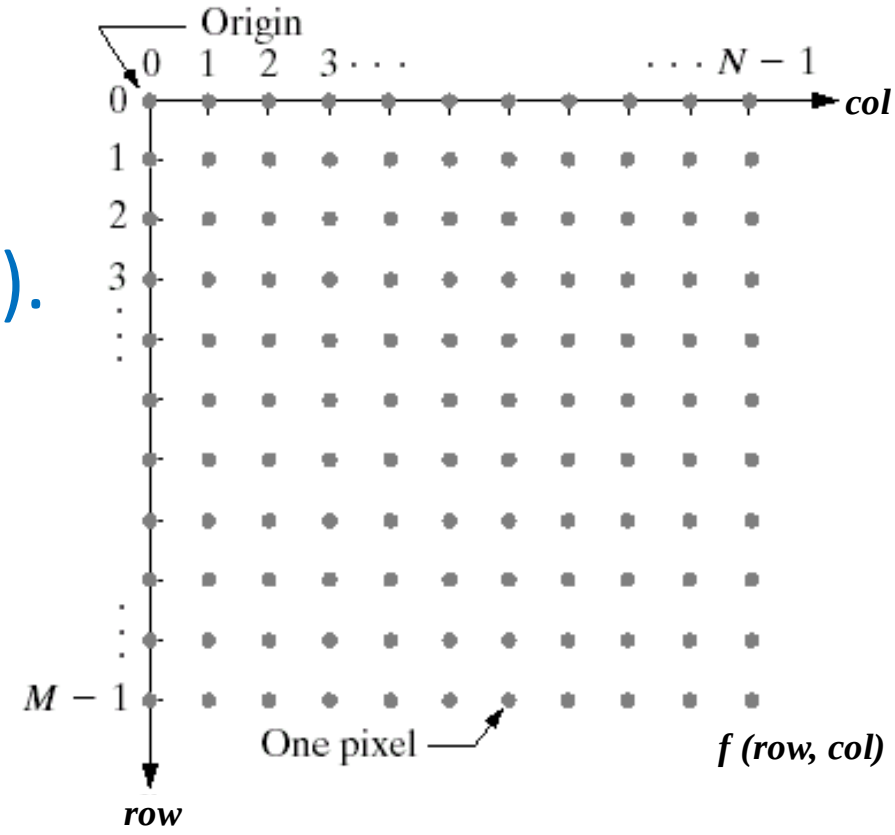
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Image Representation

A digital image is composed of M rows and N columns of pixels each storing a value.

Pixel values are most often grey levels in the range 0-255(black-white).

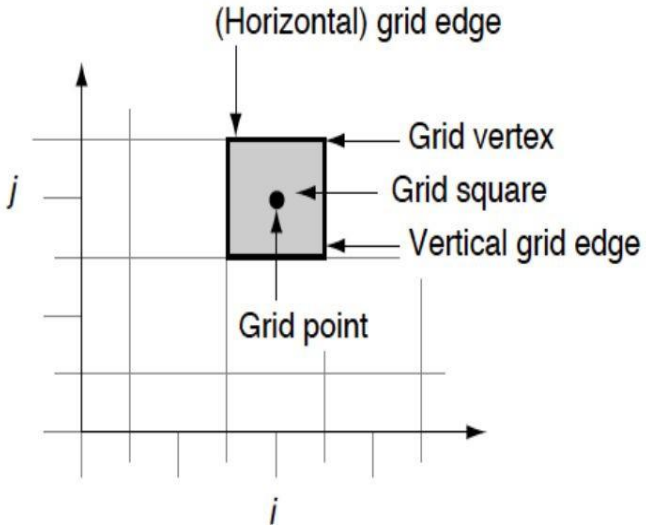
Images can easily be represented as matrices.



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Image Representation

- An image P is a function defined on a (finite) rectangular subset G also called (2D) grid, and an element of G is called a pixel (picture element).
- P assigns a value of P(p) to each $p \in G$.



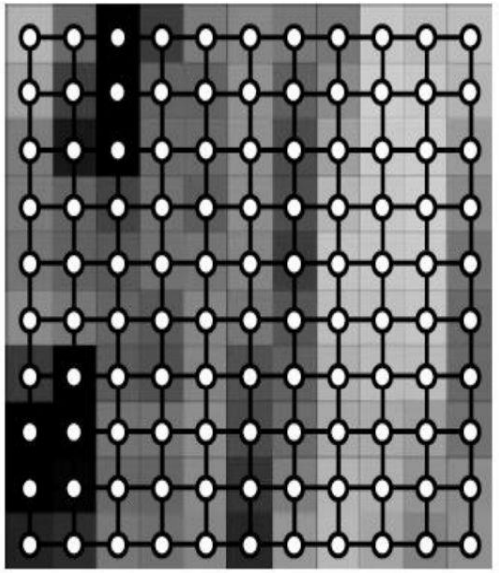
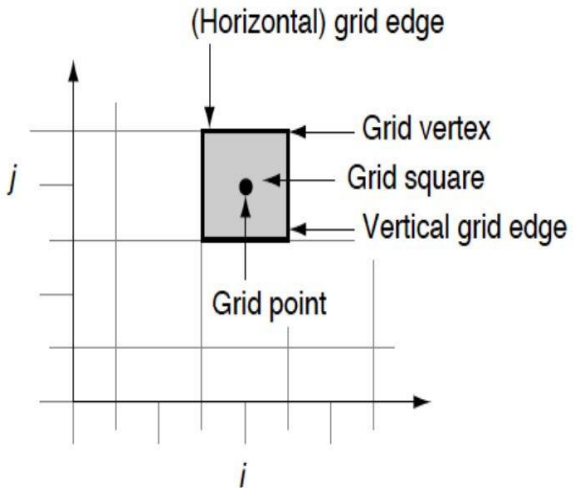


Image Representation

- Pictures are not only sampled.
- They are also quantized
  - they may have only a finite number of possible values
  - i.e., 0 to 255, 0-1, …



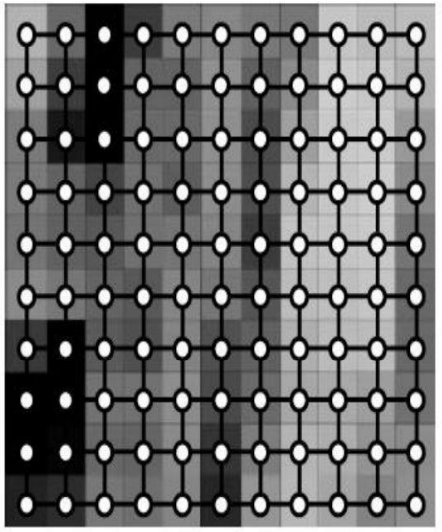
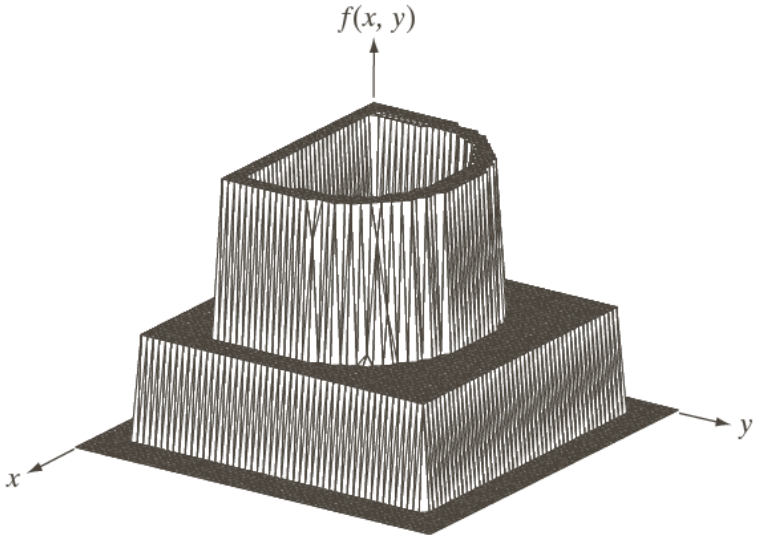
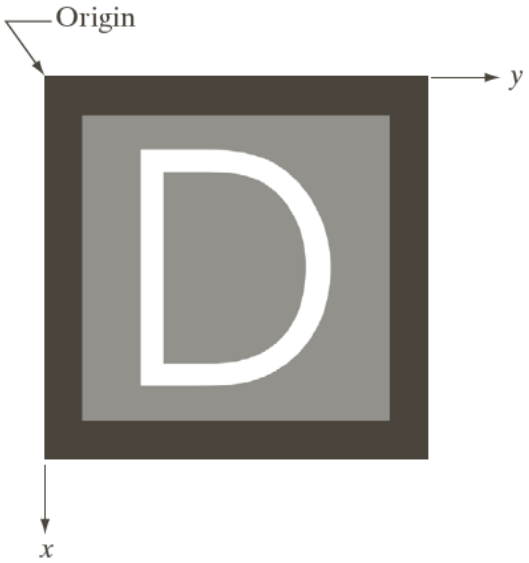
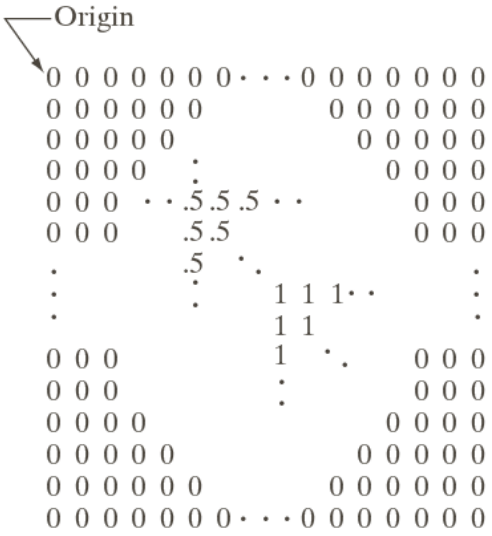


Image Representation







Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Image Representation

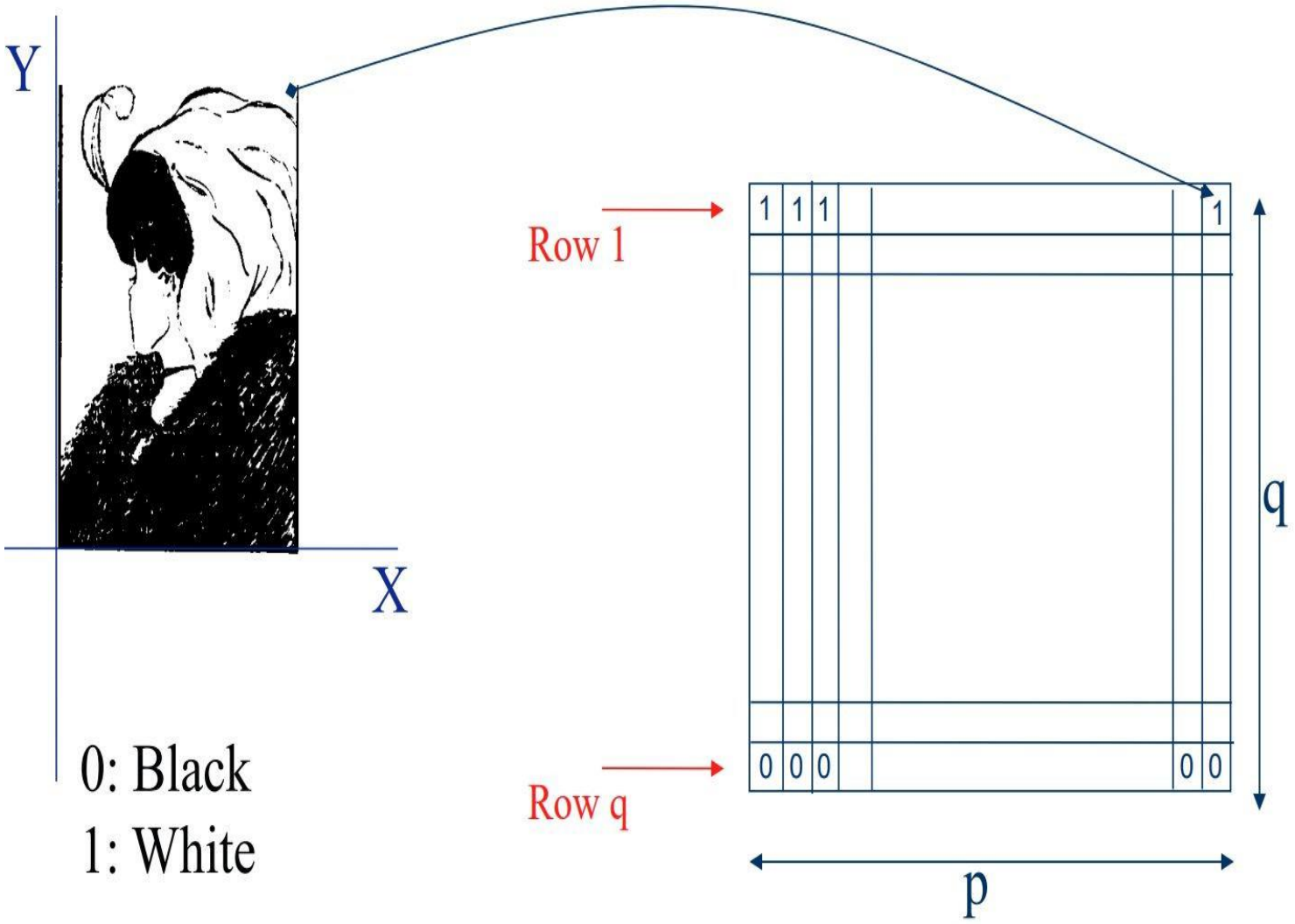
- Binary
- Grayscale
- Color



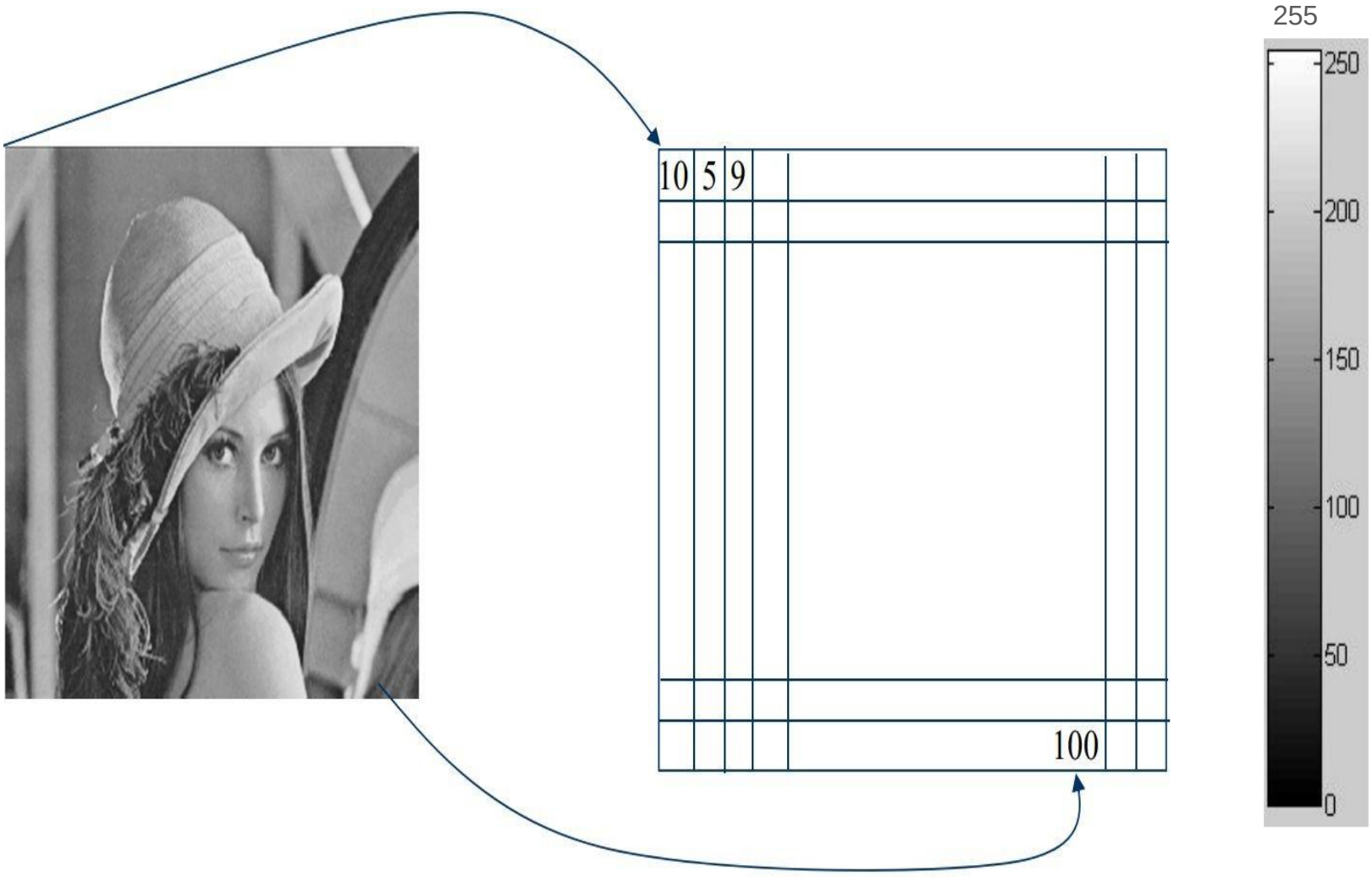




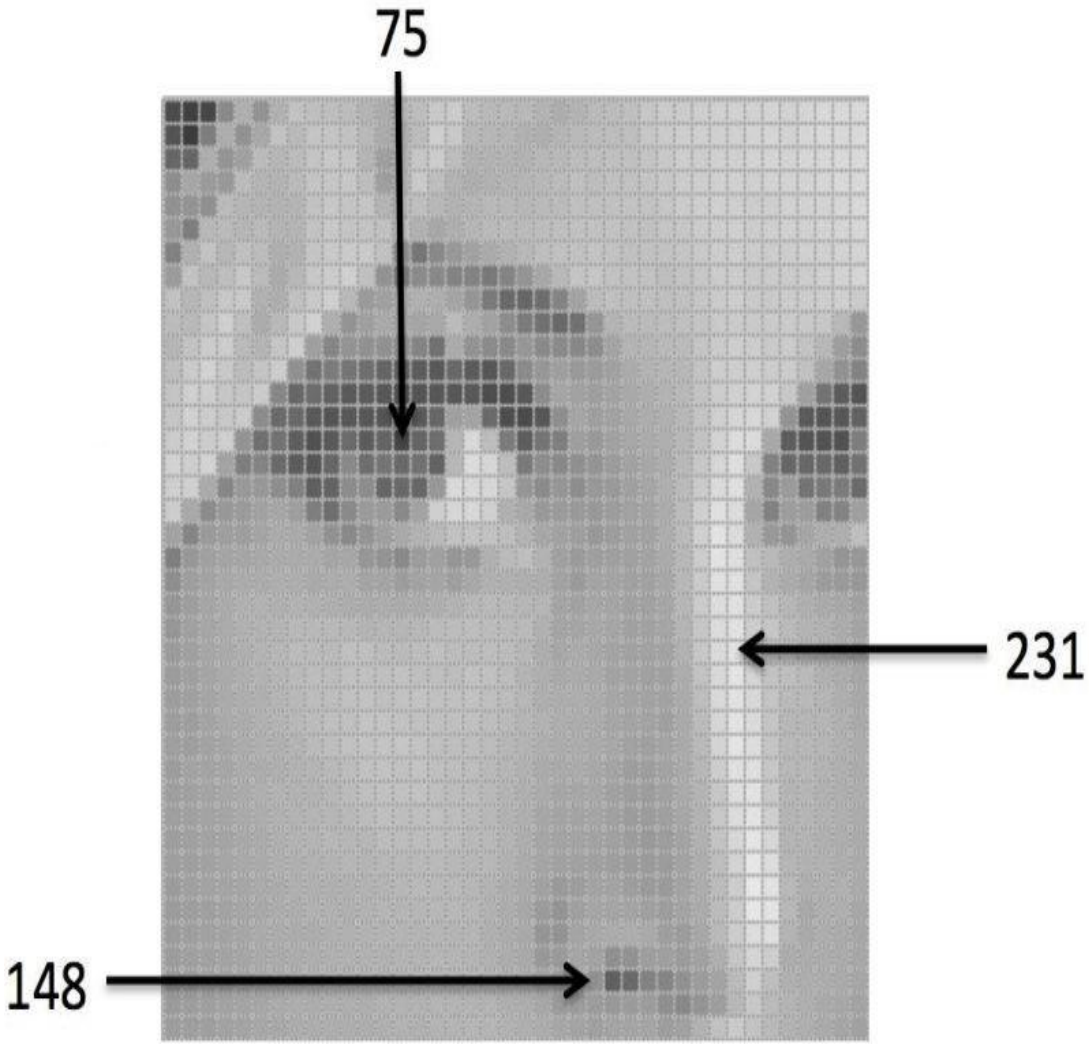
Binary Images



Grayscale Images

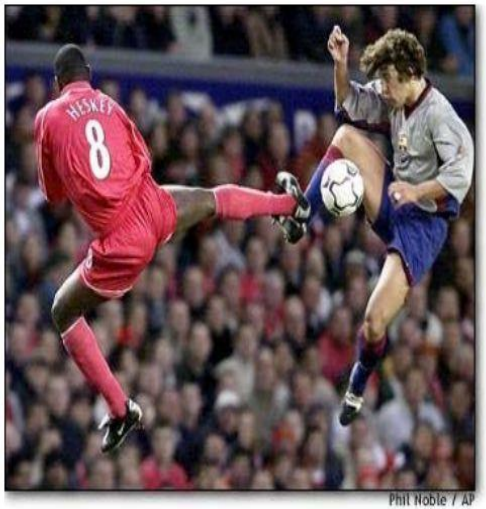


Grayscale Images



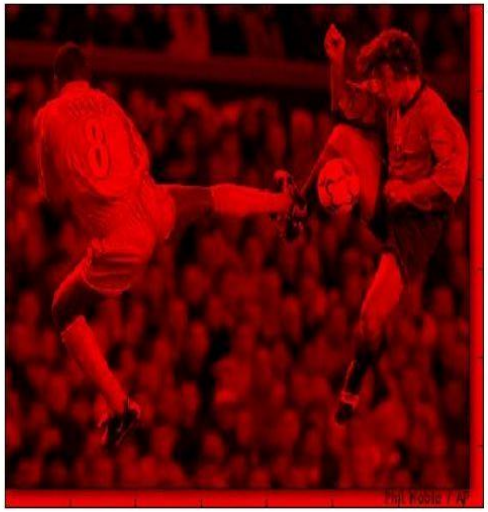
- Each pixel has a uniform color.
- More pixels allows capturing very minute color variations leading to smoother and detailed image.

Color Images

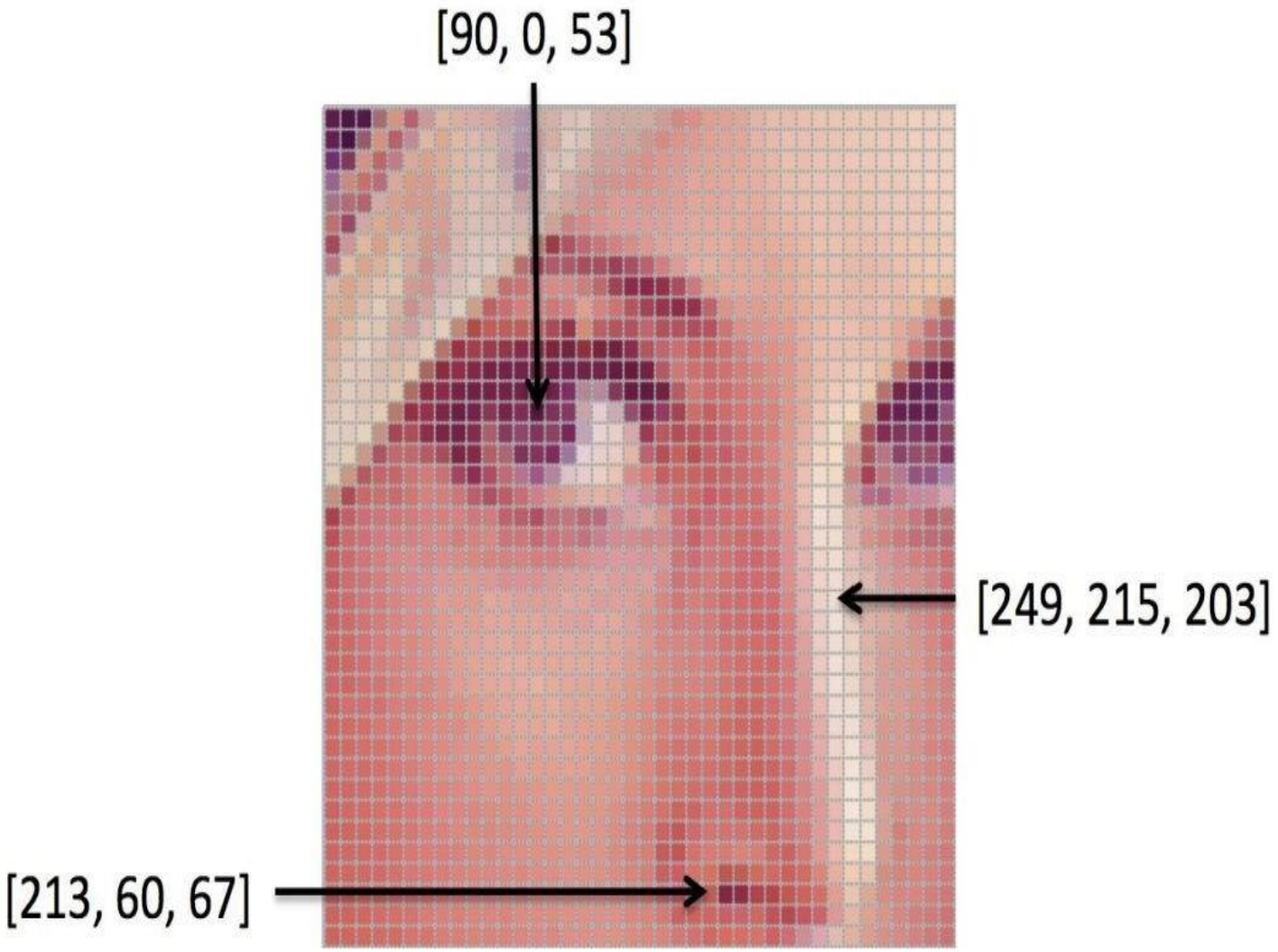








Color Images



Spatial Resolution

The spatial resolution of an image is determined by how sampling was carried out.

Spatial resolution simply refers to the smallest discernible detail in an image.

Vision specialists will often talk about pixel size.

Graphic designers will talk about *dots per inch* (DPI).



Spatial Resolution

- Sampling along the x and y axes of the image decides the resolution of the image.
  - More the sampling levels - more the resolution, better image quality, more details captured.

- Trade-off: details vs storage.



Spatial Resolution


1024


512


256


128


64

32

512×512 image is generated by deleting every other row and column from the 1024×1024 image.

The original image is of size 1024×1024.

Other images are the results of subsampling the 1024×1024 image.

Subsampling is done by deleting appropriate number of rows and colouns from the original image.

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Spatial Resolution



1024×1024

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Spatial Resolution



Virtually impossible to point out difference comparing with previous image.

512×512

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Spatial Resolution



Very slight fine checkerboard patterns in the border between flower petals and the black background.

256 × 256

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Spatial Resolution



Checkerboard patterns are increasing in the border between flower petals and the black background.

128×128

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Spatial Resolution

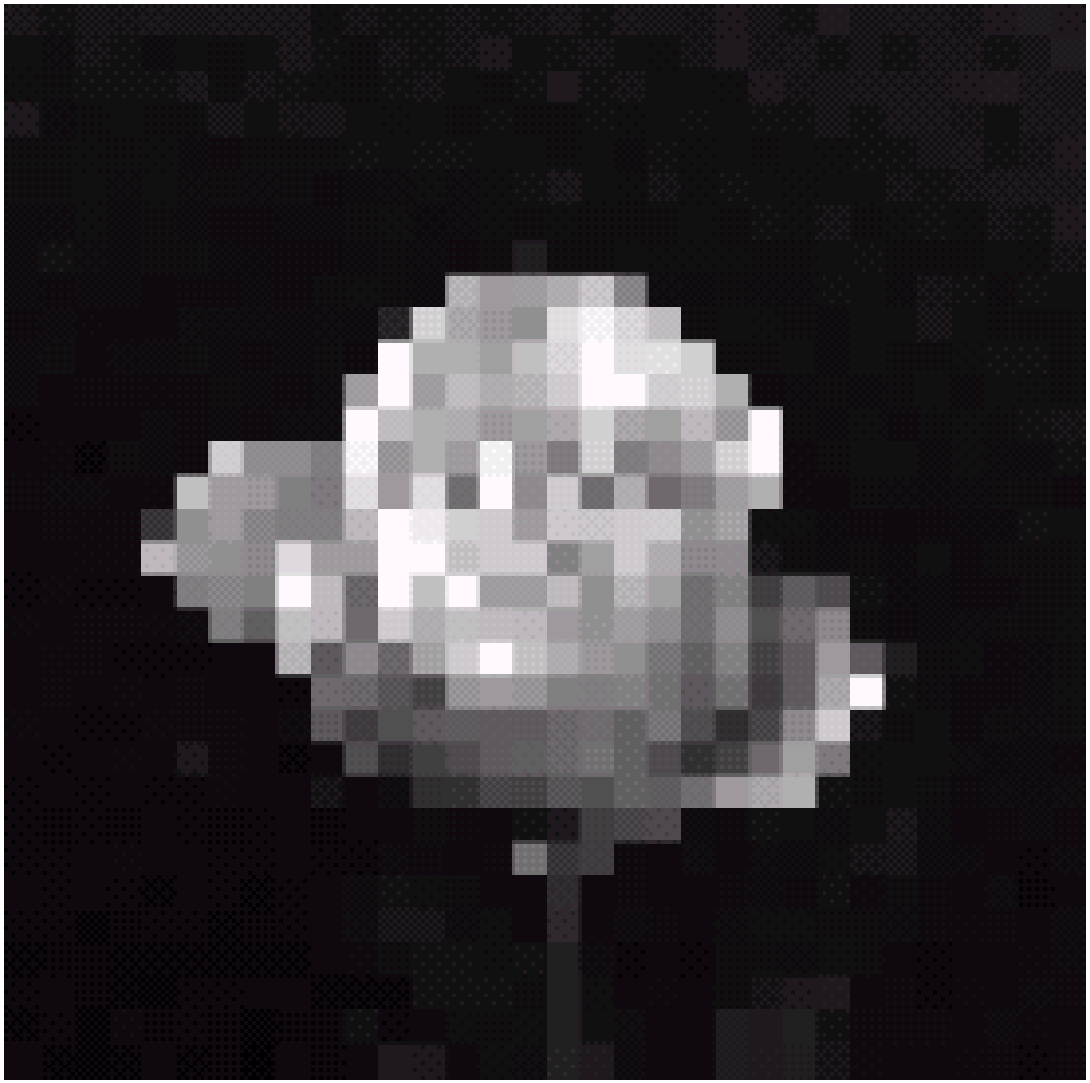


Prominent checkerboard patterns in the border between flower petals and the black background.

64×64

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Spatial Resolution



Very prominent checkerboard patterns in the border between flower petals and the black background.

32 × 32

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Intensity Level Resolution

Intensity level resolution refers to the number of intensity levels used to represent the image.

The more intensity levels used, the finer the level of detail discernable in an image.

Intensity level resolution is usually given in terms of the number of bits used to store each intensity level.

Number of Bits	Number of Intensity Levels	Examples
1	2	0, 1
2	4	00, 01, 10, 11
4	16	0000, 0101, 1111
8	256	00110011, 01010101
16	65,536	1010101010101010

Intensity Level Resolution

- Quantization of the intensity levels decides leads to fixing the levels of intensity used
  - More intensity levels - more detailed images.

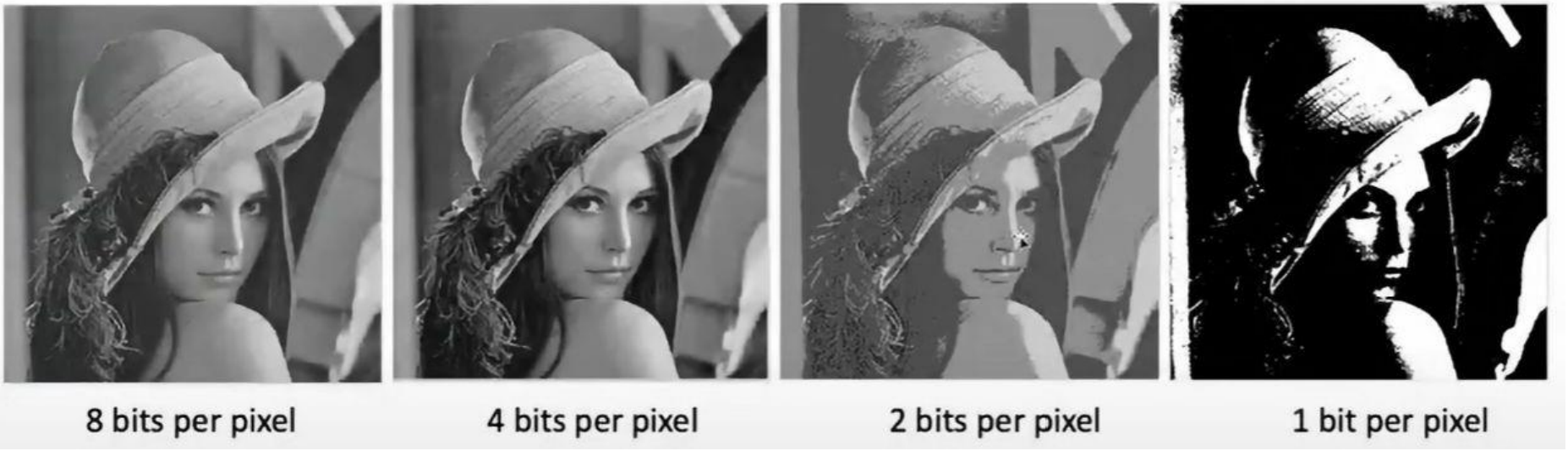
- Trade-off: details vs storage.



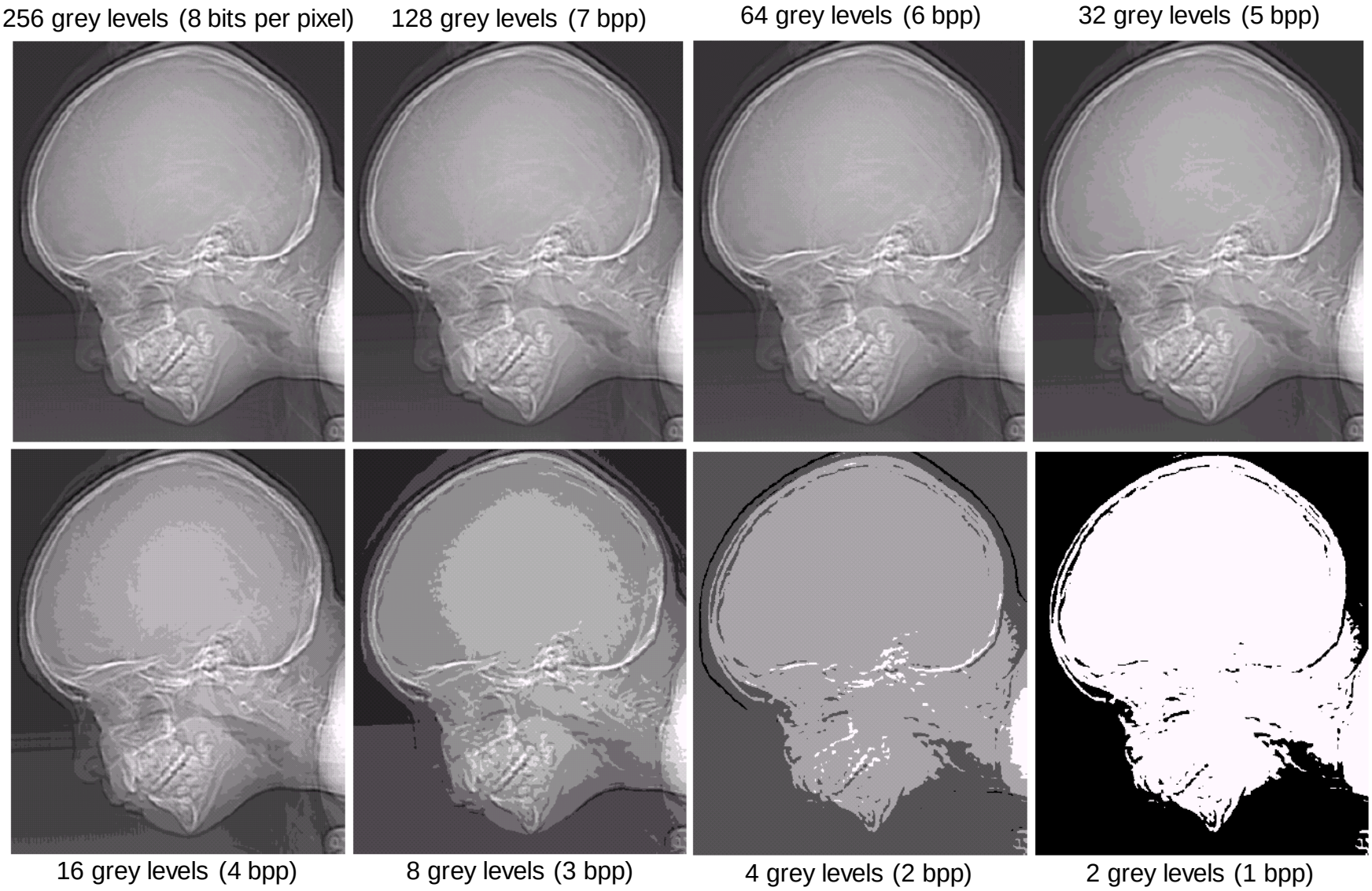
Original (256 colors) | 8 colors | 4 colors

Intensity Level Resolution

- 8 bits to represent pixel intensity - 0 to 255 intensity levels



Intensity Level Resolution



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Intensity Level Resolution



Spatial resolution 452×374 is fixed for all the images.

8 bpp

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Intensity Level Resolution



Virtually identical with previous image.

7 bpp

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Intensity Level Resolution



Virtually identical with previous image.

6 bpp

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Intensity Level Resolution



Suffering by false contouring effect.

This effect is caused by the use of an insufficient number of grey levels in smooth areas of a digital image (particularly in the skull).

5 bpp

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Intensity Level Resolution



Suffering by false contouring effect.

4 bpp

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Intensity Level Resolution

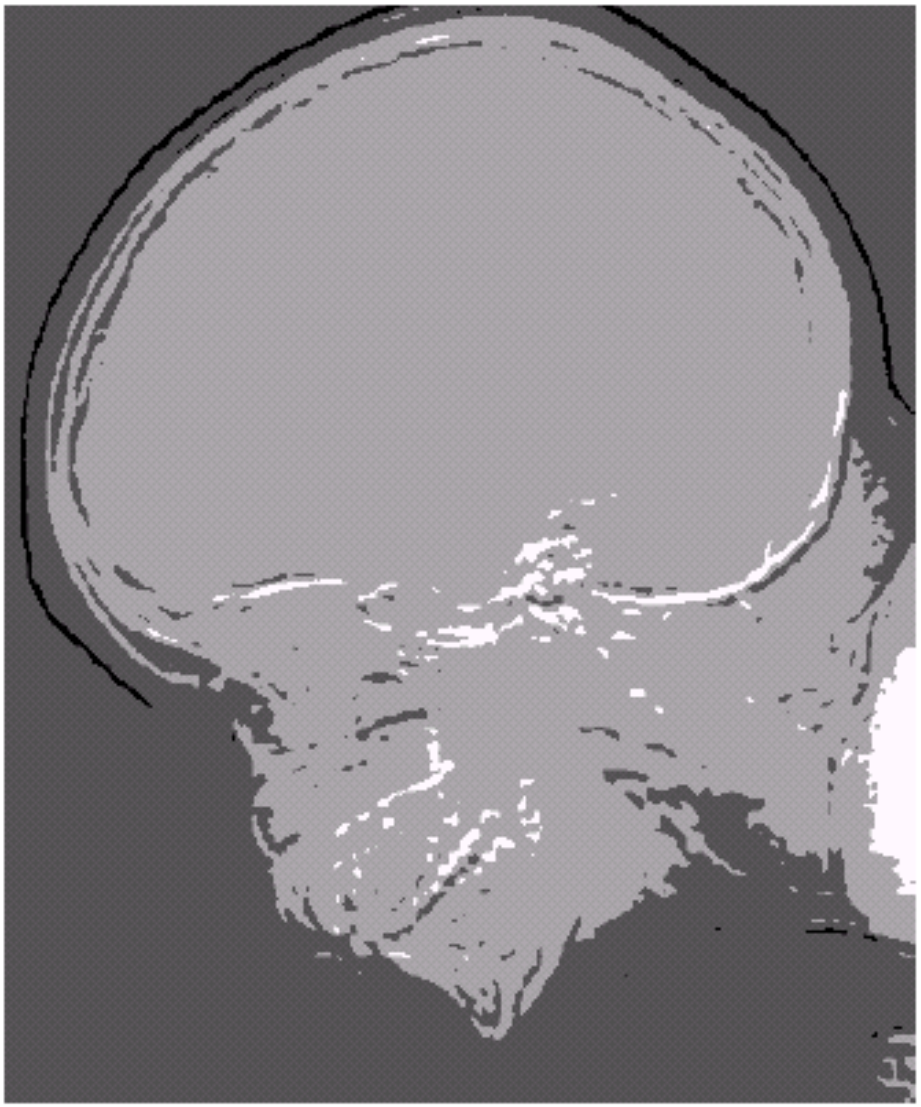


Suffering by false contouring effect.

3 bpp

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Intensity Level Resolution



Suffering by false contouring effect.

2 bpp

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Intensity Level Resolution



Suffering by false contouring effect.

1 bpp

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Thumb Rule

As a very rough rule of thumb, and assuming powers of 2 for convenience, images of size 256×256 pixels and 64 gray levels are about the smallest images, that can be expected to be reasonably free of objectionable sampling checker boards and false contouring.

Resolution: How Much is Enough?

The big question with resolution is always *how much is enough*?

This depends on what is in the image and what you would like to do with it.

Key questions include

Can you see what you need to see within the image?

Resolution: How Much is Enough?





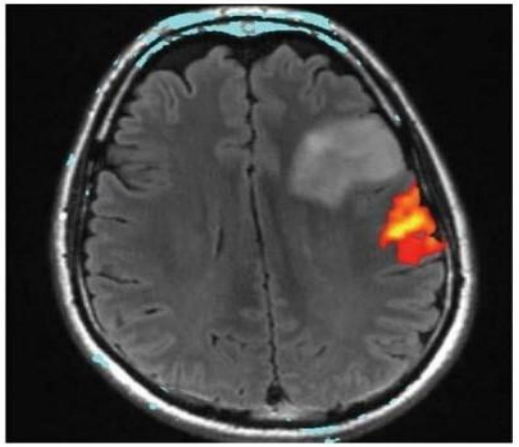
The picture on the right is fine for counting the number of cars, but not for reading the number plate.

Resolution: How Much is Enough?

- Resolution becomes especially important in some applications where very minute details are important.

- Imagine using a pixelated low resolution image to check if you have a tumor in the brain!





Effect of Spatial and Intensity Level Resolutions Simultaneously [Huang, 1965]



Image with a low level of detail



Image with a medium level of detail



Image with a relatively high level of detail

These three types of images were generated by varying spatial and intensity level resolutions.

Observers were then asked to rank them according to their subjective quality.

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Effect of Spatial and Intensity Level Resolutions Simultaneously



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Effect of Spatial and Intensity Level Resolutions Simultaneously

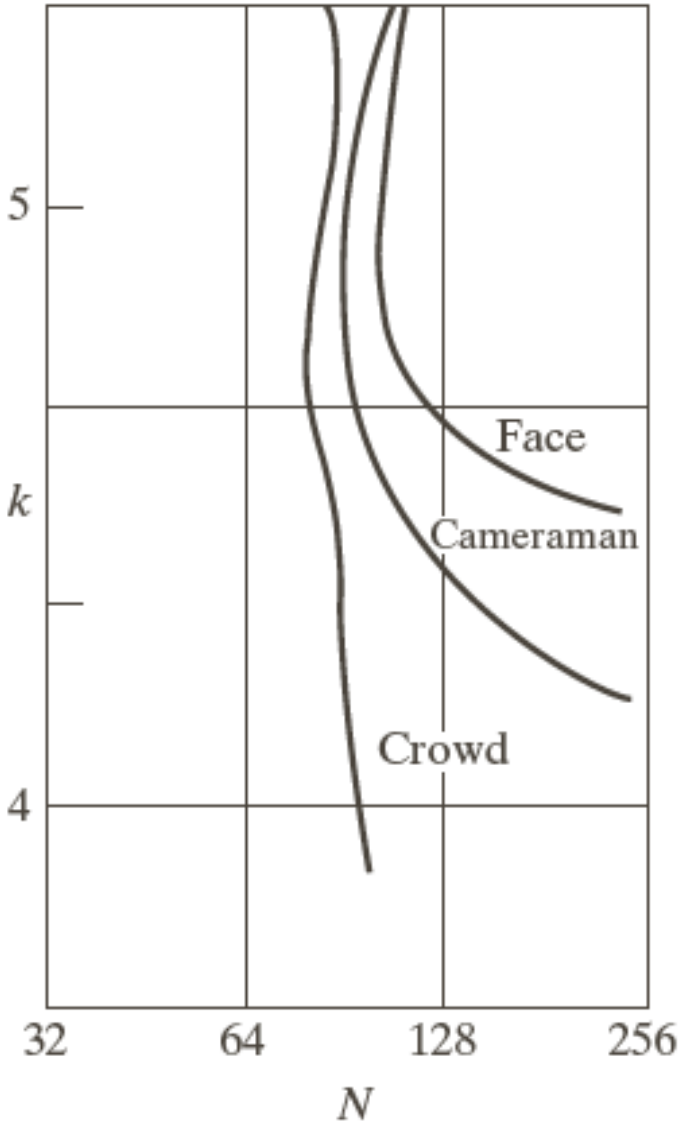


Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Effect of Spatial and Intensity Level Resolutions Simultaneously



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

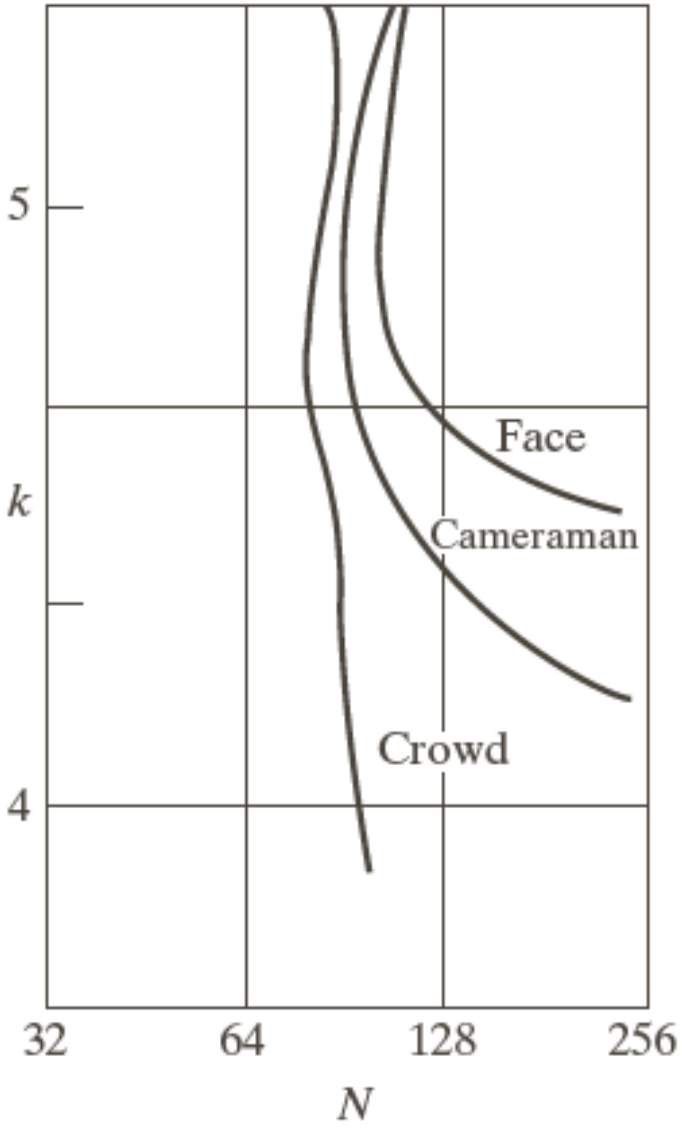


Isopreference curves

Sets of these three types of images were generated by varying *N* and *k*, and observers were then asked to rank them according to their subjective quality.

Results were summarized in the form of so-called isopreference curves in the *Nk*-plane.

This figure shows average isopreference curves representative of curves corresponding to the three types of images.

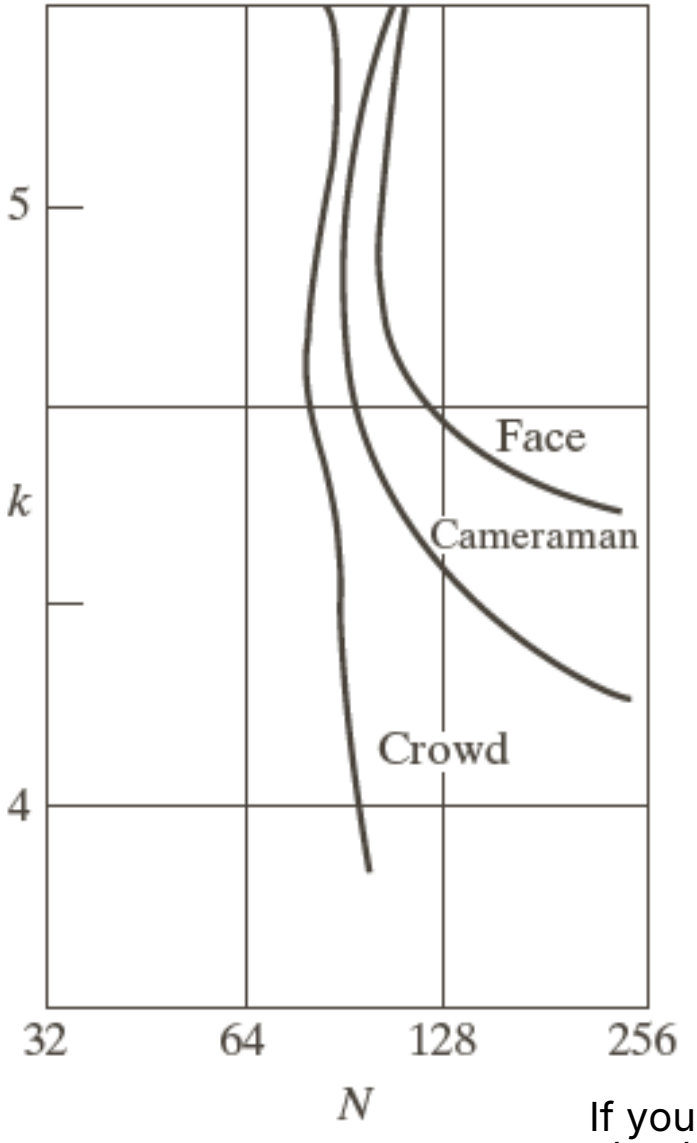


Isopreference curves

Each point in the *Nk*-plane represents an image having values of *N* and *k* equal to the coordinates of that point.

Points lying on an isopreference curve corresponds to images of equal subjective quality.

At the time of experimentation, it was found that the isopreference curves tended to shift right and upward, but their shapes in each of the three image categories were similar to those shown in this figure.



Isopreference curves

The key point of interest is that isopreference curves tend to become more vertical as the detail in the image increases.

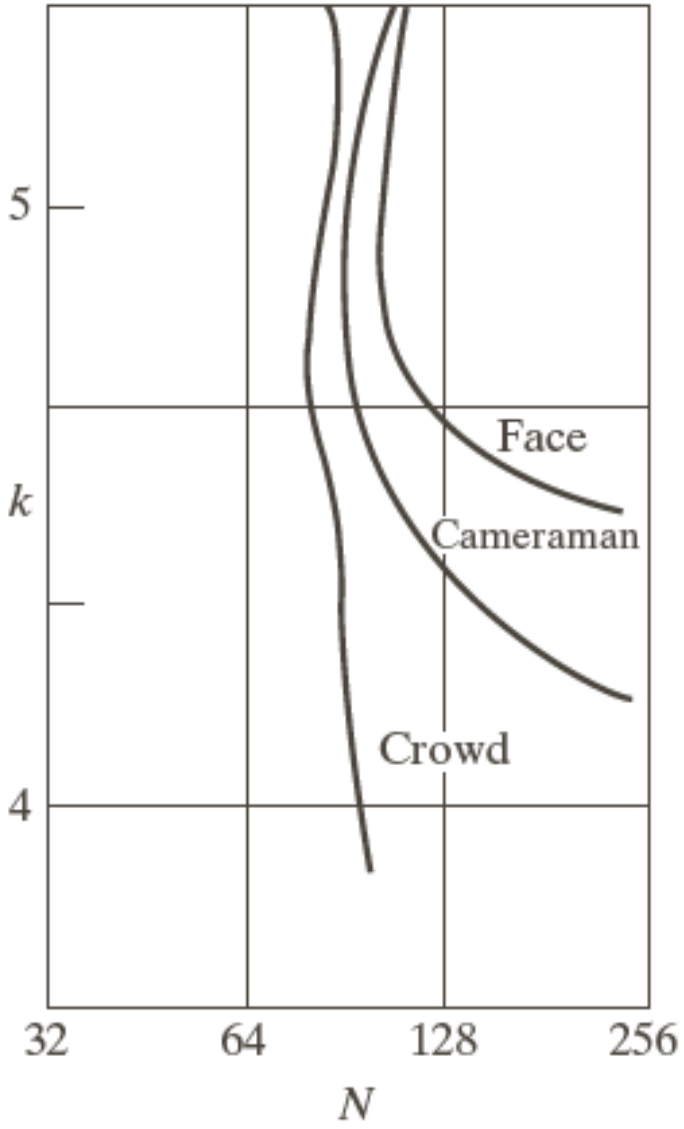
This result suggests that for images with a large amount of detail only a few gray levels may be needed.

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The curve corresponding to the crowd is nearly vertical.

This indicates that for a fixed value of N, the perceived quality for this type of image is nearly independent of k.

If you reduce N too much: → spatial details disappear, → checkerboard / aliasing-type artifacts
If you reduce k too much:→ smooth intensity transitions become bands, → false contouring



Isopreference curves

It is also of interest to note that the perceived quality in the other two image categories remained the same in some intervals in which the spatial resolution was increased, but the number of gray levels actually decreased.

The reason is that a decrease in k tends to increase the apparent contrast of an image, a visual effect that humans often perceive as improved quality in an image.